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SYSTEMS AND METHODS FOR DIRECTION OF TRAVEL

Abstract

A vehicle can include a chassis. The vehicle can also include a turntable. The turntable can be rotatably coupled to the chassis. The turntable can include a counterweight. The vehicle can also include a boom assembly. The boom assembly can have a proximal end portion coupled to the turntable and a distal end portion opposite the proximal end portion. The vehicle can also include a camera directly coupled to the counterweight. The camera can have a field of view extending away from the boom assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application is a continuation application of U.S. patent application Ser. No. 18/602,784, filed Mar. 12, 2024, which claims the benefit of and priority to both U.S. Provisional Patent Application No. 63/451,817, filed on Mar. 13, 2023, and U.S. Provisional Patent Application No. 63/451,823, filed on Mar. 13, 2023. The entireties of all of these applications are incorporated by reference herein.

BACKGROUND

[0002] The present disclosure relates to boom lifting devices.

SUMMARY

[0003] One embodiment relates to a vehicle. The vehicle can include a chassis. The vehicle can also include a turntable. The turntable can be movably coupled with the chassis. The vehicle can also include a first camera and a second camera. The vehicle can also include a control system. The control system can include one or more memory devices. The one or more memory devices can store instructions thereon. The instruction can, when executed by one or more processors, cause the one or more processors to receive first data indicating movement of the vehicle. The instructions can also cause the one or more processors to generate a first user interface including image data from the first camera. The instructions can also cause the one or more processors to in response to the first data indicating the movement of the vehicle, generate a second user interface including image data from the second camera.

[0004] One embodiment relates to a control system of a lift device. The control system can be in communication with a first camera of the lift device and a second camera of the lift device. The control system can include one or more memory devices. The one or more memory devices can store instructions thereon. The instruction can, when executed by one or more processors, cause the one or more processors to receive first data indicating movement of the lift device. The instructions can also cause the one or more processors to generate a first user interface including image data from the first camera. The instructions can also cause the one or more processors to in response to the first data indicating the movement of the lift device, generate a second user interface including image data from the second camera.

[0005] One embodiment relates to a lift device. The lift device can include a chassis. The lift device can also include a turntable. The turntable can be movably coupled with the chassis. The lift device can also include a first camera and a second camera. The lift device can also include a control system. The control system can include one or more memory devices. The one or more memory devices can store instructions thereon. The instructions can, when executed by one or more processors, cause the one or more processors to receive, via a first user interface, a request to initiate a registration process for the first camera and the second camera. The first camera and the second camera can be connected to a network that includes the lift device. The instructions can also cause the one or more processors to detect, responsive to receipt of the request, the first camera and the second camera. The instructions can also cause the one or more processors to determine, using information pertaining to the first camera and the second camera, that the first camera was previously absent from the network. The instructions can also cause the one or more processors to generate, for the first camera, an identifier for use in the network. The instructions can also cause the one or more processors to generate a second user interface to include a window to register the first camera. The window can include one or more selectable regions configured to receive input pertaining to the registration process for the first camera and the second camera. The instructions

can also cause the one or more processors to detect, responsive to an interaction with the second user interface, an indication that identifies a position of the first camera relative to the lift device. [0006] One embodiment relates to a vehicle. The vehicle can include a chassis. The vehicle can also include a turntable. The turntable can be rotatably coupled to the chassis. The turntable can include a counterweight. The vehicle can also include a boom assembly. The boom assembly can have a proximal end portion coupled to the turntable and a distal end portion opposite the proximal end portion. The vehicle can also include a camera directly coupled to the counterweight. The camera can have a field of view extending away from the boom assembly.

[0007] One embodiment relates to a vehicle. The vehicle can include a chassis. The vehicle can also include a turntable rotatably coupled to the chassis. The vehicle can also include a boom assembly having a proximal end portion coupled to the turntable and a distal end portion opposite the proximal end portion. The vehicle can also include a platform coupled to the distal end portion of the boom assembly. The platform can support an operator. An obstructed area rearward of the platform can be obstructed from view from the platform by the turntable. The vehicle can also include a camera coupled to the turntable. The camera can have a field of view including at least a portion of the obstructed area. The vehicle can also include a display coupled to the platform. The display can display image data from the camera.

[0008] One embodiment relates to a vehicle. The vehicle can include a first component. The vehicle can also include a second component rotatably coupled to the first component. The vehicle can also include an assembly having a first portion coupled to the second component and a second portion opposite the first portion. The vehicle can also include a camera coupled to the second component. The camera can have a field of view extending away from the assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

[0010] FIG. 1 is a perspective view of a fully electric boom, according to an exemplary embodiment.

[0011] FIG. 2 is a perspective view of a portion of a base assembly of the fully electric boom of FIG. 1, according to an exemplary embodiment.

[0012] FIG. 3 is a perspective view of a portion of a base assembly of the fully electric boom of FIG. 1, according to an exemplary embodiment.

[0013] FIG. 4 is a block diagram of a system including the electric boom illustrated in FIG. 1 and a control system, according to an exemplary embodiment.

[0014] FIG. 5 is a block diagram of components included in the electric boom illustrated in FIG. 1, according to an exemplary embodiment.

[0015] FIG. 6 is an aerial view of the electric boom illustrated in FIG. 1, according to an exemplary embodiment.

[0016] FIG. 7 is an aerial view of the electric boom illustrated in FIG. 1, according to an exemplary embodiment.

[0017] FIG. 8 is an aerial view of the electric boom illustrated in FIG. 1, according to an exemplary embodiment.

[0018] FIG. 9 is a flow chart illustrating a method of using control logic, according to an exemplary embodiment.

[0019] FIG. 10 is a block diagram illustrating components, according to an exemplary embodiment.

[0020] FIG. **11** is a flow chart illustrating a method of locating cameras disposed on the fully electric boom illustrated in FIG. **1**, according to an exemplary embodiment.

[0021] FIG. **12** is a user interface generated by a control system included in the system of FIG. **4**, according to an exemplary embodiment.

[0022] FIG. **13** is a user interface generated by the control system included in the system of FIG. **4**, according to an exemplary embodiment.

[0023] FIG. **14** is a user interface generated by a control system included in the system of FIG. **4**, according to an exemplary embodiment.

[0024] FIG. **15** is a user interface generated by a control system included in the system of FIG. **4**, according to an exemplary embodiment.

[0025] FIG. **16** is a perspective view of the electric boom illustrated in FIG. **1**, according to an exemplary embodiment.

[0026] FIG. **17** is a perspective view of the electric boom illustrated in FIG. **1**, according to an exemplary embodiment.

[0027] FIG. **18** is a perspective view of a platform assembly, according to an exemplary embodiment.

[0028] FIG. **19** is a perspective view of the electric boom illustrated in FIG. **1**, according to an exemplary embodiment.

[0029] FIG. **20** is a perspective view of the electric boom illustrated in FIG. **1** including a number of video zones, according to an exemplary embodiment.

[0030] FIG. **21** is a perspective view of the electric boom illustrated in FIG. **1** including a number of video zones, according to an exemplary embodiment.

[0031] FIG. **22** is a perspective view of the electric boom illustrated in FIG. **1** including a number of video zones and a number of proximity zones, according to an exemplary embodiment.

[0032] FIG. **23** is a perspective view of the electric boom illustrated in FIG. **1** including a number of video zones, according to an exemplary embodiment.

[0033] FIG. **24** is a perspective view of the electric boom illustrated in FIG. **1** including a number of video zones, according to an exemplary embodiment.

[0034] FIG. **25** is a perspective view of a panel, according to an exemplary embodiment.

[0035] FIG. **26** is a perspective view of the panel illustrated in FIG. **25**, according to an exemplary embodiment.

[0036] FIG. **27** is a perspective view of the panel illustrated in FIG. **25**, according to an exemplary embodiment.

[0037] FIG. **28** is a perspective view of the panel illustrated in FIG. **25**, according to an exemplary embodiment.

[0038] FIG. **29** is a perspective view of an assembly, according to an exemplary embodiment.

[0039] FIG. **30** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0040] FIG. **31** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0041] FIG. **32** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0042] FIG. **33** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0043] FIG. **34** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0044] FIG. **35** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0045] FIG. **36** is a perspective view of the electric boom illustrated in FIG. **1** including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0046] FIG. 37 is a perspective view of the electric boom illustrated in FIG. 1 including a number of proximity zones and a number of video zones, according to an exemplary embodiment.

[0047] FIG. 38 is a perspective view of the electric boom illustrated in FIG. 1, according to an exemplary embodiment.

[0048] FIG. 39 is a perspective view of the electric boom illustrated in FIG. 1, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0049] Before turning to the figures, which illustrate the exemplary embodiments in detail, it should be understood that the present application is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology is for the purpose of description only and should not be regarded as limiting.

Overview

[0050] Referring generally to the FIGURES, a fully electric boom is shown, according to various exemplary embodiments. The fully electric boom includes a platform assembly, a lift assembly, and a base assembly. The base assembly supports the lift assembly and the platform assembly. The platform assembly is positioned at a top end of the lift assembly. The lift assembly can include one or more articulated arms that are hingedly coupled with each other. The one or more articulated arms are configured to be driven to pivot by electric linear actuators. As the articulated arms pivot, the lift assembly increases or decreases in height, thereby raising and lowering the platform assembly.

[0051] The base assembly include a turn table and a base. The turn table is rotatably coupled with the base. The lift assembly is rotatably coupled with the turn table. The turn table is driven to rotate relative to the base assembly by an electric turn table motor. The base assembly also includes one or more tractive elements. The tractive elements each include an electric motor configured to drive the corresponding tractive element. The tractive elements can be independently driven by the corresponding electric motor. The base assembly includes a steering system. The steering system includes an electric linear actuator that extends and retracts. An end of a rod of the electric linear actuator is fixedly coupled with an end of one or more arcuate/curved steering members. The one or more arcuate/curved steering members are pivotally coupled with a steering knuckle of the tractive element. An end of the housing of the electric linear actuator is pivotally/rotatably coupled with the base. The electric linear actuator can be extended or retracted to pivot the corresponding tractive element for a turn.

[0052] The electric turntable motor is rotatably coupled with a gear box. The gear box receives rotational kinetic energy from the electric turntable motor and outputs rotational kinetic energy with a higher torque. Depending on the orientation/configuration of the electric turntable motor and the gear box, a transmission can be used to transfer the rotational kinetic energy into an axis that is substantially parallel to an axis that a ring gear of the turntable rotates about (e.g., using bevel gears). In other embodiments, the transmission or the gear box includes a worm and the ring gear is a worm gear.

[0053] The fully electric boom can include energy storage devices (e.g., batteries). Any of the motors, electric linear actuators, etc., of the fully electric boom can receive electrical power from the energy storage devices. A controller is configured to receive user inputs from one or more human machine interfaces and operate any of the motors, electric linear actuators, etc., of the fully electric boom. The controller may generate control signals for any of the electric motors, electric linear actuators, etc. The controller can also monitor feedback (e.g., voltage feedback, current feedback, etc.) from any of the electric linear actuators, electric motors, etc.

[0054] The fully electric boom can include a boom arm and a jib arm. The boom arm is configured to lower and raise by one or more electric linear actuators. The jib arm is coupled to an end of the boom arm and is configured to rotate and pivot the platform assembly. The jib arm includes a platform rotator that pivotally couples the platform assembly with the jib arm.

[0055] The platform rotator pivotally couples the platform assembly with the jib arm. The platform rotator can include a barrel formed by two portions that couple with the platform assembly. The barrel can fixedly couple with one or more structural support members that protrude outwards from the platform assembly.

[0056] The barrel is supported on either side (e.g., an upper side and a lower side) by structural support members that extend from the end of the jib arm. The barrel can slidably interface with the structural support members. The barrel and the structural support members are configured to rotatably or pivotally couple with each other.

[0057] The platform rotator includes an electric motor, a brake, and one or more gear boxes. The electric motor is configured to drive the one or more gear boxes to pivot the barrel relative to the structural support members that support the barrel. The one or more gear boxes can be reduction gear boxes that increase the output torque provided to the barrel. The brake can be an electric brake that transitions between an activated state and a deactivated state. When in the activated state, the brake facilitates restricting relative rotation between the barrel and the structural support members that support the barrel.

[0058] The electric motor and the brake of the platform rotator can receive electrical energy from the batteries of the electric boom. The electric motor and the brake can be operated by a controller in response to the controller receiving a user input from a human machine interface. The controller can operate the electric motor to pivot/rotate the platform assembly in either direction. The controller may transition the brake into the activated state to facilitate locking a current angular position of the platform assembly.

[0059] The fully electric boom can include a boom arm and a jib arm. The boom arm can include lower members, upper members, and two upright members. The lower members are pivotally coupled with the turntable at a lower end. The lower members are pivotally coupled at an upper end with a first one of the upright members. The upper members are pivotally coupled at a lower end with the first upright member, and pivotally coupled with the second upright member at the upper end. The jib arm is pivotally coupled with the second upright member. The lower members can pivot about the lower end to raise and lower the first upright member. The upper members can pivot about their lower ends to raise and lower the second upright member and the jib arm.

[0060] An electric linear actuator is pivotally coupled at a lower end with one of the lower members. The electric linear actuator is mounted with a trunnion mount to one of the upper members at an upper end. The trunnion mount includes a collar that surrounds a cylinder of the electric linear actuator. The collar can be a single-piece clamping collar or a two-piece clamping collar. The collar includes protrusions that extend radially outwards and pivotally couple with apertures of one of the upper members. The trunnion mount facilitates using an electric linear actuator with a longer overall stroke length. The electric linear actuator can be operated to pivot the upper members about their bottom ends, and thereby raise/lower the second upright member.

[0061] Another electric linear actuator can be pivotally coupled at a lower end with the turntable and pivotally coupled with one of the lower members with a trunnion mount. The electric linear actuator can be operated to extend and retract to pivot the lower members about their lower ends.

[0062] The electric boom further includes an axle lock out system configured to selectively limit rotation of the axle assemblies relative to the chassis. The axle assemblies are configured to rotate relative to the chassis about a longitudinal axis. A pair of electric linear actuators (i.e., axle actuators) are coupled to the chassis on opposite sides of the longitudinal axis. The electric linear actuators extend downward from the chassis to engage the corresponding axle assembly. During a driving mode of operation, the axle actuators permit free rotation of the axle assembly. In some embodiments, during the driving mode, the axle actuators are held in a retracted configuration to permit the axle to rotate through a first range of motion without contacting the axle actuators. In other embodiments, during the driving mode, the axle actuators engage the axle assemblies, but are configured to extend and retract freely (e.g., with minimal resistance) such that the axle assembly

can rotate through the first range of motion.

[0063] During an operating mode, the axle actuators limit rotation of the axle assemblies to a second range of motion smaller than the first range of motion. In embodiments where the axle actuators are held in the retracted position during the driving mode, the axle actuators are extended until they engage the axle assembly. A controller may determine that the axle actuators have engaged the axle assembly in response to the current supplied to each actuator reaching a threshold current. Once the axle actuators have engaged the axle assembly, the axle actuators may lock to a fixed length. In embodiments where the axle actuators extend and retract freely during the driving mode, the axle actuators may lock to a fixed length in response to entering the operating mode.

[0064] Advantageously, the fully electric boom does not use any hydraulic systems, hydraulic pumps, engines, internal combustion engines, etc. to perform the respective functions of the various motors and actuators. All of the motors and actuators are fully electric. Other systems use electric motors to rotate pumps of various hydraulic systems. The fully electric boom facilitates a quieter, more environmentally friendly, more efficient lift device.

Electric Boom

[0065] Referring to FIG. 1, an electric lifting apparatus, an electric telehandler, an electric boom lift, a towable electric boom lift, a lift device, a fully electric boom lift, etc., shown as electric boom **10** includes a base assembly **12** (e.g., a support assembly, a drivable support assembly, a support structure, etc.), a platform assembly **16** (e.g., a platform, a terrace, etc.), and a lift assembly **14** (e.g., a boom lift assembly, a lifting apparatus, an articulated arm, a scissors lift, a boom assembly, etc.). If electric boom **10** is a telehandler, platform assembly **16** can be replaced with a fork apparatus, a bucket apparatus, a material lifting apparatus, a mechanical lifting apparatus attachment, etc. Electric boom **10** includes a front end (e.g., a forward facing end, a front portion, a front, etc.), shown as front **62**, and a rear end (e.g., a rearward facing end, a back portion, a back, a rear, etc.), shown as rear **60**. Lift assembly **14** is configured to elevate platform assembly **16** in an upwards direction **46** relative to base assembly **12**. Lift assembly **14** is also configured to translate platform assembly **16** in a downwards direction **48**. Lift assembly **14** is also configured to translate platform assembly **16** in either a forwards direction **50** or a rearwards direction **51**. Lift assembly **14** generally facilitates performing a lifting function to raise and lower platform assembly **16**, as well as movement of platform assembly **16** in various directions.

[0066] In some embodiments, electric boom **10** may be outfitted and/or implemented for use in different applications. Accordingly, the electric boom **10** can include a common vehicle that can be configured for a variety of different uses simply by selecting an appropriate application kit **80**. By way of example, the electric boom **10** may be configured as a refuse vehicle, a concrete mixer, a fire fighting vehicle, an airport fire fighting vehicle, a lift device (e.g., a boom lift, a scissor lift, a telehandler, a vertical lift, etc.), a crane, a tow truck, a military vehicle, a delivery vehicle, a mail vehicle, a boom truck, a plow truck, a farming machine or vehicle, a construction machine or vehicle, a coach bus, a school bus, a semi-truck, a passenger or work vehicle (e.g., a sedan, a SUV, a truck, a van, etc.), and/or still another vehicle.

[0067] Base assembly **12** defines a longitudinal axis **78** and a lateral axis **80**. Longitudinal axis **78** defines forwards direction **50** of electric boom **10** and rearwards direction **51**. Electric boom **10** is configured to translate in forwards direction **50** and to translate backwards in rearwards direction **51**. Base assembly **12** includes one or more wheels, tires, wheel assemblies, tractive elements, rotary elements, treads, etc., shown as tractive elements **82**. Tractive elements **82** are configured to rotate to drive (e.g., translate, steer, move, etc.) electric boom **10**. Tractive elements **82** can each include an electric motor **52** (e.g., electric wheel motors) configured to drive tractive elements **82** (e.g., to rotate tractive elements **82** to facilitation motion of electric boom **10**). In other embodiments, tractive elements **82** are configured to receive power (e.g., rotational mechanical energy) from electric motors **52** through a drive train (e.g., a combination of any number and configuration of a shaft, an axle, a gear reduction, a gear train, etc.). Tractive elements **82** and

electric motors **52** can facilitate a driving and/or steering function of electric boom **10**.

[0068] Platform assembly **16** is configured to provide a work area for an operator of electric boom **10** to stand/rest upon. Platform assembly **16** can be pivotally coupled to an upper end of lift assembly **14**. Electric boom **10** is configured to facilitate the operator accessing various elevated areas (e.g., lights, platforms, the sides of buildings, building scaffolding, trees, power lines, etc.). Electric boom **10** uses various electrically powered motors and electrically powered linear actuators to facilitate elevation of platform assembly **16** (e.g., relative to base assembly **12**, or to a ground surface that base assembly **12** rests upon).

[0069] Platform assembly **16** includes a base member, a base portion, a platform, a standing surface, a shelf, a work platform, a floor, a deck, etc., shown as deck **18**. Deck **18** provides a space (e.g., a floor surface) for a worker to stand upon as platform assembly **16** is raised and lowered.

[0070] Platform assembly **16** includes various members, beams, bars, guard rails, rails, railings, etc., shown as rails **22**. Rails **22** extend along substantially an entire perimeter of deck **18**. Rails **22** provide one or more members for the operator of electric boom **10** to grasp while using electric boom **10** (e.g., to grasp while operating electric boom **10** to elevate platform assembly **16**). Rails **22** can include members that are substantially horizontal to deck **18**. Rails **22** can also include vertical structural members that couple with the substantially horizontal members. The vertical structural members can extend upwards from deck **18**.

[0071] Platform assembly **16** can include a human machine interface (HMI) (e.g., a user interface), shown as HMI **20**. HMI **20** is configured to receive user inputs from the operator at platform assembly **16** to facilitate operation of electric boom **10**. HMI **20** can include any number of buttons, levers, switches, keys, etc., or any other user input device configured to receive a user input to operate electric boom **10**. HMI **20** can be supported by one or more of rails **22**.

[0072] Platform assembly **16** includes a frame **24** (e.g., structural members, support beams, a body, a structure, etc.) that extends at least partially below deck **18**. Frame **24** can be integrally formed with deck **18**. Frame **24** is configured to provide structural support for deck **18** of platform assembly **16**. Frame **24** can include any number of structural members (e.g., beams, bars, I-beams, etc.) to support deck **18**. Frame **24** couples platform assembly **16** with lift assembly **14**. Frame **24** may rotatably or pivotally coupled with lift assembly **14** to facilitate rotation of platform assembly **16** about an axis **28** (e.g., a centerline). Frame **24** can also rotatably/pivotally couple with lift assembly **14** such that frame **24** and platform assembly **16** can pivot about an axis **25** (e.g., a centerline).

[0073] Lift assembly **14** includes one or more beams, articulated arms, bars, booms, arms, support members, boom sections, cantilever beams, etc., shown as lift arms **32**. Lift arms **32** are hingedly or rotatably coupled with each other at their ends. Lift arms **32** can be hingedly or rotatably coupled to facilitate articulation of lift assembly **14** and raising/lowering of platform assembly **16**. Electric boom **10** includes a lower lift arm **32a**, a central or medial lift arm **32b**, and an upper lift arm **32c**. Lower lift arm **32a** is configured to hingedly or rotatably couple at one end with base assembly **12** to facilitate lifting (e.g., elevation) of platform assembly **16**. Lower lift arm **32a** is configured to hingedly or rotatably couple at an opposite end with medial lift arm **32b**. Likewise, medial lift arm **32b** is configured to hingedly or rotatably couple with upper lift arm **32c**. Upper lift arm **32c** can be configured to hingedly interface/couple and/or telescope with an intermediate lift arm **32d**. Upper lift arm **32c** can be referred to as “the jib” of electric boom **10**. Intermediate lift arm **32d** may extend into an inner volume of upper lift arm **32c** and extend/retract. Lower lift arm **32a** and medial lift arm **32b** may be referred to as “the boom” of electric boom **10**. Intermediate lift arm **32d** can be configured to couple (e.g., rotatably, hingedly, etc.), with platform assembly **16** to facilitate levelling of platform assembly **16**.

[0074] Lift arms **32** are driven to hinge or rotate relative to each other by electric actuators **34** (e.g., electric linear actuators, linear electric arm actuators, etc.). Electric actuators **34** can be mounted between adjacent lift arms **32** to drive adjacent lift arms **32** to hinge or pivot (e.g., rotate some

angular amount) relative to each other about pivot points **84**. Electric actuators **34** can be mounted between adjacent lift arms **32** using any of a foot bracket, a flange bracket, a clevis bracket, a trunnion bracket, etc. Electric actuators **34** are configured to extend or retract (e.g., increase in overall length, or decrease in overall length) to facilitate pivoting adjacent lift arms **32** to pivot/hinge relative to each other, thereby articulating lift arms **32** and raising or lowering platform assembly **16**.

[0075] Electric actuators **34** can be configured to extend (e.g., increase in length) to increase a value of angle **74** formed between adjacent lift arms **32**. Angle **74** can be defined between centerlines of adjacent lift arms **32** (e.g., centerlines that extend substantially through a center of lift arms **32**). For example, electric actuator **34a** is configured to extend/retract to increase/decrease angle **74a** defined between a centerline of lower lift arm **32a** and longitudinal axis **78** (angle **74a** can also be defined between the centerline of lower lift arm **32a** and a plane defined by longitudinal axis **78** and lateral axis **80**) and facilitate lifting of platform assembly **16** (e.g., moving platform assembly **16** at least partially along upwards direction **46**). Likewise, electric actuator **34b** can be configured to retract to decrease angle **74a** to facilitate lowering of platform assembly **16** (e.g., moving platform assembly **16** at least partially along downwards direction **48**). Similarly, electric actuator **34b** is configured to extend to increase angle **74b** defined between centerlines of lower lift arm **32a** and medial lift arm **32b** and facilitate elevating of platform assembly **16**. Similarly, electric actuator **34b** is configured to retract to decrease angle **74b** to facilitate lowering of platform assembly **16**. Electric actuator **34c** is similarly configured to extend/retract to increase/decrease angle **74c**, respectively, to raise/lower platform assembly **16**.

[0076] Electric actuators **34** can be mounted (e.g., rotatably coupled, pivotally coupled, etc.) to adjacent lift arms **32** at mounts **40** (e.g., mounting members, mounting portions, attachment members, attachment portions, etc.). Mounts **40** can be positioned at any position along a length of each lift arm **32**. For example, mounts **40** can be positioned at a midpoint of each lift arm **32**, and a lower end of each lift arm **32**.

[0077] Intermediate lift arm **32d** and frame **24** are configured to pivotally interface/couple at a platform rotator **30** (e.g., a rotary actuator, a rotational electric actuator, a gear box, etc.). Platform rotator **30** facilitates rotation of platform assembly **16** about axis **28** relative to intermediate lift arm **32d**. In some embodiments, platform rotator **30** is between frame **24** and upper lift arm **32c** and facilitates pivoting of platform assembly **16** relative to upper lift arm **32c**. Axis **28** extends through a central pivot point of platform rotator **30**. Intermediate lift arm **32d** is also configured to extend/retract along upper lift arm **32c**. Intermediate lift arm **32d** can also be configured to pivotally/rotatably couple with upper lift arm **32c** such that intermediate lift arm **32d** pivots/rotates about axis **25**. Intermediate lift arm **32d** can be driven to rotate/pivot about axis **25** by extension and retraction of electric actuator **34d**.

[0078] Platform assembly **16** is configured to be driven to pivot about axis **28** (e.g., rotate about axis **28** in either a clockwise or a counter-clockwise direction) by an electric motor **26** (e.g., a rotary electric actuator, a stepper motor, a platform rotator, a platform electric motor, an electric platform rotator motor, etc.). Electric motor **26** can be configured to drive frame **24** to pivot about axis **28** relative to upper lift arm **32c** (or relative to intermediate lift arm **32d**). Electric motor **26** can be configured to drive a gear train to pivot platform assembly **16** about axis **28**.

[0079] Lift assembly **14** is configured to pivotally or rotatably couple with base assembly **12**. Base assembly **12** include a rotatable base member, a rotatable platform member, a fully electric turntable, etc., shown as turntable **70**. Lift assembly **14** is configured to rotatably/pivotally couple with base assembly **12**. Turntable **70** is rotatably coupled with a base, frame, structural support member, carriage, etc., of base assembly **12**, shown as base **36**. Turntable **70** is configured to rotate or pivot relative to base **36**. Turntable **70** can pivot/rotate about central axis **42** relative to base **36**. Turntable **70** facilitates accessing various elevated and angularly offset locations at platform assembly **16**. Turntable **70** is configured to be driven to rotate or pivot relative to base **36** by an

electric motor, an electric turntable motor, an electric rotary actuator, etc., shown as turntable motor **44**. Turntable motor **44** can be configured to drive a ring gear that is rotatably coupled with base **36** to produce relative rotation of turntable **70** relative to base **36**. Lower lift arm **32a** is pivotally coupled with turntable **70** (or with a turntable member **72** of turntable **70**) such that lift assembly **14** and platform assembly **16** rotate as turntable **70** rotates about central axis **42**. In some embodiments, turntable **70** is configured to rotate a complete 360 degrees about central axis **42** relative to base **36**. In other embodiments, turntable **70** is configured to rotate an angular amount less than 360 degrees about central axis **42** relative to base **36** (e.g., 270 degrees, 120 degrees, etc.).

[0080] Base assembly **12** includes one or more energy storage devices (e.g., capacitors, batteries, Lithium-Ion batteries, Nickel Cadmium batteries, etc.), shown as batteries **64**. Batteries **64** are configured to store energy in a form (e.g., in the form of chemical energy) that can be converted into electrical energy for the various electric motors and electric actuators of electric boom **10**. Batteries **64** can be stored within base **36**. Electric boom **10** includes a controller **38** configured to operate any of the electric motors, electric actuators, etc., of electric boom **10**. Controller **38** can be configured to receive sensory input information from various sensors of electric boom **10**, user inputs from HMI **20** (or any other user input device such as a key-start or a push-button start), etc. Controller **38** can be configured to generate control signals for the various electric motors, electric actuators, etc., of electric boom **10** to operate any of the electric motors, electric actuators, electrically powered movers, etc., of electric boom **10**. Batteries **64** are configured to power any of the electrical motors, sensors, actuators, electric linear actuators, electrical devices, electrical movers, stepper motors, etc., of electric boom **10**. Base assembly **12** can include a power circuit including any necessary transformers, resistors, transistors, thermistors, capacitors, etc., to provide appropriate power (e.g., electrical energy with appropriate current and/or appropriate voltage) to any of the electric motors, electric actuators, sensors, electrical devices, etc., of electric boom **10**.

[0081] Batteries **64** are configured to deliver power to electric motors **52** to drive tractive elements **82**. A rear set of tractive elements **82** can be configured to pivot to steer electric boom **10**. In other embodiments, a front set of tractive elements **82** are configured to pivot to steer electric boom **10**. In still other embodiments, both the front and the rear set of tractive elements **82** are configured to pivot (e.g., independently) to steer electric boom **10**.

[0082] Base assembly **12** can include one or more laterally extending frame members (e.g., laterally extending structural members) and one or more longitudinally extending frame members (e.g., longitudinally extending structural members).

[0083] Base assembly **12** includes a steering system **150**. Steering system **150** is configured to drive tractive elements **82** to pivot for a turn of electric boom **10**. Steering system **150** can be configured to pivot tractive elements **82** in pairs (e.g., to pivot a front pair of tractive elements **82**), or can be configured to pivot tractive elements **82** independently (e.g., four-wheel steering for tight-turns).

[0084] Base assembly **12** can include an HMI **21** (e.g., a user interface, a user input device, a display screen, etc.). In some embodiments, HMI **21** is coupled with base **36**. In other embodiments, HMI **21** is positioned on turntable **70**. HMI **21** can be positioned on any side or surface of base assembly **12** (e.g., on the front **62** of base **36**, on the rear **60** of base **36**, etc.).

[0085] Controller **38** can receive user inputs from HMI **21** and/or HMI **20** and operate any of electric motors **52**, electric actuators **34**, electric motor **26**, turntable motor **44**, and axle actuators **56** to operate electric boom **10**. For example, controller **38** may receive a user input from HMI **21** or HMI **20** to elevate platform assembly **16** and may operate electric actuators **34** to raise or lower platform assembly **16**. Likewise, controller **38** can receive a user input from HMI **21** or HMI **20** to rotate turntable **70** about axis **42** relative to base **36** and can operate turntable motor **44** to rotate turntable **70** based on the user input. Controller **38** can also receive a user input from HMI **21** or HMI **20** to drive or steer electric boom **10** and can operate electric motors **52** and electric actuator(s) to drive and steer tractive elements **82**. Controller **38** operates any of electric motors **52**,

electric actuators **34**, electric actuators **122**, electric motor **26**, and turntable motor **44** by generating control signals and providing the control signals to the various controllable elements to perform requested operations of electric boom **10**.

[0086] Controller **38** can receive sensor inputs from sensors of electric boom **10**. Sensors can include proximity sensors, distance sensors, position sensors, etc. Sensors can include any safety sensors that measure relative distance between electric boom **10** and objects. Sensors can include sensors that monitor an approximate elevation of lift assembly **14**. In other embodiments, sensors can include temperature sensors configured to measure a temperature of batteries **64** to determine a condition of batteries **64**. Controller **38** can also receive feedback from any of electric motors **52**, electric actuators **34**, electric motor **26**, turntable motor **44**, and axle actuators **56**. In some embodiments, the feedback information includes voltage or current indicative of a position (e.g., linear position, degree of extension, angular position, etc.) of any of the controllable elements, a speed (e.g., a speed of extension, a speed of rotation, etc.) of any of the controllable elements, etc. [0087] For example, the feedback received from turntable motor **44** can indicate a current angular position of turntable motor **44**. Controller **38** can use any of the feedback from electric motors **52**, electric actuators **34**, electric motor **26**, turntable motor **44**, and axle actuators **56** to track, monitor, etc., angular or linear position of any of the controllable elements. In some embodiments, the feedback is received from a sensor associated with each of the controllable elements. For example, a position sensor can be mounted to each of electric actuators **34** to monitor a degree of extension or retraction of electric actuators **34**. Controller **38** can use any of the feedback information to monitor operations of electric boom **10** and to generate control signals for the controllable elements.

[0088] Controller **38** can monitor whether any of the controllable elements of electric boom **10** are operating properly based on the feedback received from the controllable elements. For example, controller **38** may receive feedback from any of the controllable elements (e.g., linear electric actuators of lift assembly **14**, turntable motor **44**, platform rotator **30**, etc.) and detect failure of any of the controllable elements based on the received feedback. In some embodiments, controller **38** notifies an operator regarding any failed controllable elements. For example, if controller **38** determines that electric actuator **34a** is not operating properly, controller **38** can notify the operator by providing a message to the operator through HMI **20** and/or HMI **21**.

[0089] Controller **38** can also monitor sensory information measured by sensors to determine if any of the controllable elements are not operating properly. For example, if the sensory information from sensors indicates that a particular one of electric actuators (or any of the controllable elements) has not extended an expected amount, controller **38** can determine that the particular one of electric actuators is not operating properly. Controller **38** can provide a notification to the operator through HMI **20** and/or HMI **21** regarding any detected failures of the controllable elements (e.g., any of the electric motors, any of the electric actuators, etc.).

[0090] Electric boom **10** can also include one or more weight sensors configured to measure a load applied to platform assembly **16** (or forks, lifting apparatus, buckets, etc., if electric boom **10** is a telehandler). Controller **38** can receive sensor measurements from the weight sensors indicating the load applied to platform assembly **16**. Controller **38** can generate control signals for any of the controllable elements (e.g., electric actuators, electric motors, electric rotary actuators, etc.) of electric boom **10** based on the load applied to platform assembly **16**. For example, if the load applied to platform assembly **16** is greater than a threshold value, controller **38** can restrict operation of lift assembly **14**.

[0091] Controller **38** can include a communications interface. The communication interface can facilitate communications between controller **38** and external systems, devices, sensors, etc. (e.g., sensors, HMI **20**, HMI **21**, electric motors **52**, electric actuators **34**, electric motor **26**, turntable motor **44**, axle actuators **56**, etc.) for allowing user control, monitoring, and adjustment to any of the communicably connected devices, sensors, systems, primary movers, etc. The communications

interface can also facilitate communications between controller **38** and HMI **21** and/or HMI **20** (e.g., a touch screen, a display screen, a personal computer, etc.) or with a network.

[0092] Controller **38** can include a processing circuit. The processing circuit can include a processor and memory. The processing circuit can be communicably connected to the communications interface such that the processing circuit and the various components thereof can send and receive data via the communications interface. The processor can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

[0093] Memory (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory can be or include volatile memory or non-volatile memory. Memory can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to some embodiments, memory is communicably connected to the processor via the processing circuit and includes computer code for executing (e.g., by the processing circuit and/or the processor) one or more processes described herein.

[0094] Electric motors **52**, electric actuators **34**, electric motor **26**, turntable motor **44**, and axle actuators **56** can receive electrical power from batteries **64** to perform any of their respective operations. Controller **38** can generate control signals for any of the controllable elements to perform their respective operation in response to receiving a user input from HMI **21** and/or HMI **20**. When the controllable elements receive the control signals from controller **38**, the controllable elements use the electrical power provided by batteries **64** to perform their respective operations.

[0095] Controller **38** can receive a user input from HMI **21** and/or HMI **20** to raise or lower platform assembly **16** and generates control signals to cause electric actuators **34** to raise or lower platform assembly **16** (e.g., to raise or lower platform assembly **16** the amount desired/input by the user/operator). Likewise, controller **38** can receive a user input from HMI **21** and/or HMI **20** to rotate turntable **70** and can generate control signals for turntable motor **44** to rotate turntable **70** (e.g., to rotate turntable **70** the desired amount as input by the user/operator).

[0096] Controller **38** can also generate and provide control signals to turntable brakes to restrict rotation of turntable **70**. The turntable brake can receive electric power from batteries **64** and actuates between an activated state and a deactivated state to restrict and allow rotation of a ring gear, respectively. Controller **38** may operate turntable motor **44** in response to receiving a user input from HMI **20** and/or HMI **21**. Controller **38** can receive feedback from turntable motor **44** indicating an angular position of turntable motor **44** or an angular speed of turntable motor **44**. In some embodiments, controller **38** receives sensory information from a turntable sensor that indicates an angular position of turntable **70**. In some embodiments, the user input received from HMI **20** and/or HMI **21** indicates a direction of rotation of turntable **70**. Controller **38** generates control signals and provides the control signals to turntable motor **44** to operate turntable motor **44** to rotate turntable **70** in the desired direction of rotation.

[0097] Controller **38** can also provide control signals to an electric actuator to raise/lower the boom arm of lift assembly **14**. The electric actuator can use electric power from batteries **64** to operate an electric motor. Controller **38** can operate the electric actuator to extend or retract by operating the electric motor to operate in a forwards direction or a backwards direction. Controller **38** can operate the electric motor to cause the electric actuator to extend in response to receiving a user input from HMI **20** and/or HMI **21** to raise platform assembly **16**. Likewise, controller **38** can operate the electric motor to cause the electric actuator to retract in response to receiving a user input from HMI **20** and/or HMI **21** to lower platform assembly **16**.

[0098] Controller **38** can also generate and provide control signals to electric motor **26** and/or

brakes. Controller **38** can generate and provide the control signals to electric motor **26** to operate electric motor **26** in either direction, thereby pivoting platform assembly **16** about axis **28** in either direction. Controller **38** can operate electric motor **26** in response to receiving user inputs from HMI **20** and/or HMI **21**. For example, an operator can provide controller **38** with a user input to pivot/rotate platform assembly **16** in a clockwise direction at HMI **20** and/or HMI **21** (e.g., by pressing a button, pulling a lever, moving a joy-stick, etc.). Controller **38** can operate electric motor **26** as long as the user input from HMI **20** and/or HMI **21** is received. In some embodiments, controller **38** can receive a user input from HMI **20** and/or HMI **21** to lock platform assembly **16** at a current angular position. Controller **38** can generate and provide control signals to the brakes to lock platform assembly **16** (e.g., to activate brake **26**) at a current angular position in response to receiving the user input from HMI **20** and/or HMI **21**. Likewise, controller **38** can receive a user input from HMI **20** and/or HMI **21** to de-activate the brakes. Controller **38** can generate and provide control signals to the brakes to transition the brakes into the de-activated state in response to receiving a user input from HMI **20** and/or HMI **21**.

[0099] Referring now to FIGS. 2-3, base assembly **12** includes a longitudinally extending frame member **54** (e.g., a rigid member, a structural support member, an axle, a base, a frame, a carriage, etc.). Longitudinally extending frame member **54** provides structural support for turntable **70** as well as tractive elements **82**. Longitudinally extending frame member **54** is pivotally coupled with lateral frame members **110** (e.g., axles, frame members, beams, bars, etc.) at opposite longitudinal ends of longitudinally extending frame member **54**. For example, lateral frame members **110** may be pivotally coupled with longitudinally extending frame member **54** at a front end and a rear end of longitudinally extending frame member **54**. Lateral frame members **110** can be configured to pivot about a pivot joint **58**. Pivot joint **58** can include a pin and a receiving portion (e.g., a bore, an aperture, etc.). The pin of pivot joint **58** is coupled to one of lateral frame member **110** (e.g., a front lateral frame member **110** or a rear lateral frame member **110**) or longitudinally extending frame member **54** and the receiving portion is coupled to the other of longitudinally extending frame member **54** and lateral frame member **110**. For example, the pin may be coupled with longitudinally extending frame member **54** and the receiving portion can be coupled with one of lateral frame members **110** (e.g., integrally formed with the front lateral frame member **110**).

[0100] In some embodiments, longitudinally extending frame member **54** and lateral frame members **110** are integrally formed or coupled (e.g., fastened, welded, riveted, etc.) to define base **36**. In still other embodiments, base **36** is integrally formed with longitudinally extending frame member **54** and/or lateral frame members **110**. In still other embodiments, base **36** is coupled with longitudinally extending frame member **54** and/or lateral frame members **110**.

[0101] Base assembly **12** includes one or more axle actuators **56** (e.g., electric linear actuators, electric axle actuators, electric levelling actuators, etc.). Axle actuators **56** can be linear actuators configured to receive power from batteries **64**. Axle actuators **56** can be configured to extend or retract to contact a top surface of a corresponding one of lateral frame members **110**.

[0102] When axle actuators **56** extend, an end of a rod of electric levelling actuators can contact the surface of lateral frame member **110** and prevent relative rotation between lateral frame member **110** and longitudinally extending frame member **54**. In this way, the relative rotation/pivoting between lateral frame member **110** and longitudinally extending frame member **54** can be locked (e.g., to prevent rolling of longitudinally extending frame member **54** relative to lateral frame members **110** during operation of lift assembly **14**). Axle actuators **56** can receive power from batteries **64**. Axle actuators **56** receive control signals from controller **38**. Axle actuators **56** receive electrical power (e.g., to extend or retract) from batteries **64**.

Control System

[0103] FIG. 4 depicts a block diagram of a system **400** including the electric boom illustrated in FIG. 1 and a control system, according to an exemplary embodiment. Each system and/or component of the system **400** can include one or more processors, memory, network interfaces,

communication interfaces, and/or user interfaces. The memory can store programming logic that, when executed by the processor, controls the operation of the corresponding computing system or device. The memory can also store data in databases. The network interfaces can allow the systems and/or components of the system **400** to communicate wirelessly. The communication interfaces can include wired and/or wireless communication interfaces and the systems and/or components of the system **400** can be connected via the communication interfaces. The various components in the system **400** can be implemented via hardware (e.g., circuitry), software (e.g., executable code), or any combination thereof. Systems, devices, and components in FIG. **4** can be added, deleted, integrated, separated, and/or rearranged.

[0104] The system **400** can include the electric boom **10**, at least one network **480**, and at least one user device **485**. The electric boom **10** and/or a component thereof can communicate, via the network **480**, with the user device **485**. The electric boom **10** can be at least one of the vehicles described herein. The electric boom **10** can include at least one control system **405**, at least one sensor **455**, at least one camera **460**, at least one display device **465**, at least one input/output (I/O) device **470**, and at least one Ethernet Switch **475**. The components of the electric boom **10** and/or the system **400** can be electrically coupled with one another. For example, the control system **405** can be electrically coupled with the sensor **455**. The components of the electric boom **10** and/or the components of the system **400** can also communicate with, interact with, and/or otherwise interface with one another via a controller area network (CAN). For example, the control system **405** can communicate, via a CAN, with the display device **465**.

[0105] The sensors **455** can be and/or include an accelerometer, a tachometer, a speedometer, a GPS device/sensor, a temperature sensor, a voltmeter, an ammeter, a radar sensor, a pressure sensor, a tactile sensor, a photodetector, a motion sensor, a proximity sensor, and/or among other possible sensors and/or devices. For example, the sensors **455** can be a tactile sensor. The sensors **455** can be and/or include any additional sensor and/or type of sensor described herein. The sensors **455** can also collect and/or otherwise provide information similar to that of any sensor described herein. The sensor **455** can provide operational data associated with the electric boom **10** to the control system **405**. The operational data associated with the electric boom **10** can be and/or include at least one of a speed of the electric boom **10**, an acceleration of the electric boom **10**, a location of the electric boom **10**, an operator input (e.g., an input provided by an operator of the electric boom **10**), movement and/or actions performed and/or pertaining to the controllable elements of the electric boom **10** and/or an environmental condition of the electric boom **10**. For example, the operational data can include information pertaining to operation of the turntable **70** (e.g., the turntable **70** rotated and/or otherwise moved).

[0106] The cameras **460** can be and/or include at least one of a video camera that captures video, a camera that captures images and/or among other possible optical instruments and/or optical devices that can capture, record, produce and/or otherwise provide video. The cameras **460** can also include audio devices. For example the cameras **460** can include at least one of a speaker, a microphone, a headphone, and/or among other possible audio and/or sound devices. The camera **460** can provide video data to the control system **405**. The video data can be and/or include video feeds, images, recordings, audio files, audio signals, field of views, and/or any other possible information that can be captured, produced and/or otherwise provided by the cameras **460**. The cameras **460** can be placed, located, situated, positioned, coupled and/or otherwise disposed on various components and/or locations on the electric boom **10**. For example, a camera **460** can be disposed on the platform assembly **16**.

[0107] The display devices **465** can be and/or include a screen, a monitor, a visual display device, a television, a video display, a liquid crystal display (LCD), a light emitting diode (LED) display, an infotainment system, a mobile device, and/or among other possible displays and/or devices. For example, the display device **465** can be an LCD. The display device **465** can generate, produce, provide and/or otherwise display a user interface. For example, the display device **465** can display

a user interface that includes video feeds provided by the cameras **460**.

[0108] The I/O device **470** can be and/or include a joystick, a button, a toggle switch, a handle, a lever, a steering wheel, a key, a keypad, a console, a mouse, a keyboard, a knob, a dial, a vehicle shifter, a smart phone, a computer, a wearable device, and/or among other possible I/O devices. For example, the I/O device **470** can be a joystick. An operator of the electric boom **10** can interact with, interface with and/or otherwise engage with the I/O device **470**. For example, the operator can select a vehicle mode button on a joystick (e.g., the I/O device **470**). The operating engaging with the I/O device **470** can result in the I/O device **470** communicating with the control system **405** and/or a component of the system **400**.

[0109] The Ethernet switch **475** can be and/or include at least one of a network switch, a modem, a router, a switch and/or among any other possible device and/or component that connect components to one another over a network. For example, the Ethernet switch **475** can connect, over the network **480**, the turntable **70** with the base assembly **12**.

[0110] The network **480** can be and/or include a local area network (LAN), wide area network (WAN), telephone network (such as the Public Switched Telephone Network (PSTN)), wireless link, intranet, the Internet, a cellular network and/or combinations thereof. The user device **485** can be and/or include at least one of a mobile computing device, a desktop computer, a smartphone, a tablet, a smart watch, a smart sensor and/or any other device configured to facilitate providing, receiving, displaying and/or otherwise interacting with content (e.g., webpages, mobile applications, etc.).

[0111] An operator of the user device **485** can perform various actions and/or access various types of information. The information can be provided over the network **480** (e.g., the Internet, LAN, WAN, cellular, etc.). Similarly, the user device **485** can be perform similar functionality to that of the display device **465**. The user device **485** can include an application to receive and display information and to receive user interactions with the content. For example, the application can be a web browser and/or a mobile application.

[0112] The control system **405** can include at least one processing circuit **410**, at least one monitor component **425**, at least one video manager **430**, at least on interface generator **435**, at least one registration manager **440**, at least one network interface **445**, and at least one controller **450**. The control system **405** and/or a component thereof can be and/or include components that can be used to control various aspect of the electric boom **10**. For example, the control system **405** can control movement of the electric boom **10**.

[0113] The processing circuit **410** can include at least one processor **415** and memory **420**. Memory **420** can be one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing and/or facilitating the various processes described herein. For example, memory **420** can store instructions and the instructions can cause the processor **415** to perform functionality similar to that of the control system **405** and/or a component thereof. Memory **420** can be or include non-transient volatile memory, non-volatile memory, and non-transitory computer storage media. Memory **420** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described herein. Memory **420** can be communicably coupled with the processor **415**. Memory **420** can also be electrically coupled with the processor **415**. Memory **420** can include computer code or instructions for executing one or more processes described herein. The processor **415** can be implemented as one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), a group of processing components, and/or other suitable electronic processing components.

[0114] The network interface **445** can be and/or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with sensors, devices, systems, etc., of electric boom **10** or other external systems or devices (e.g., the user device **485**). The network interface **445** can be direct

(e.g., local wired or wireless communications) or via a communications network (e.g., the network **480**). For example, the network interface **445** can include an Ethernet card and port for sending and receiving data via an Ethernet-based communications link or network. The network interface **445** can also include a Wi-Fi transceiver for communicating via a wireless communications network (e.g., the network **480**). The network interface **445** can include a power line communications interface. The network interface **445** can include an Ethernet interface, a USB interface, a serial communications interface, and/or a parallel communications interface. The network interface **445** can interface with, interact with and/or otherwise communicate with at least one of the user device **485**, the sensors **455**, the camera **460**, the display device **465**, the I/O device **470**, the Ethernet switch **475**, and/or any other possible component of the electric boom **10**.

[0115] The controller **450** can control various controllable elements described herein pertaining to the electric boom **10**. The controller **450** can be and/or include the controller **38**. The controller **450** can receive a request to move and/or otherwise control a controllable element of the electric boom **10**. For example, the controller **450** can receive the request responsive to the operator of the electric boom **10** selecting an icon displayed on a user interface. The controller **450** can, responsive to receiving the request, provide control signals to the controllable element associated with the request. For example, the request can be a request to rotate the turntable **70** and the controller **450** can send control signals to the turntable motor **44** causing the turntable motor **44** to rotate the turntable **70**.

[0116] The monitor component **425** can determine at least one of a position, an orientation, a movement and/or any other possible action performed and/or taken by the electric boom **10**. For example, the monitor component **425** can determine that the turntable **70** has rotated with respect to the base assembly **12**. The monitor component **425** can receive, from the sensors **455**, a set of data pertaining to an operation of the electric boom **10**. For example, the monitor component **425** can receive a set of data pertaining to the turntable **70** rotating. The set of data provided by the sensors **455** can be and/or include information generated by the controllable elements of the electric boom **10**. For example, the set of data can include information indicating a position, a placement and/or a degree of extension or retraction of any of the controllable elements described herein. For example, the set of data can indicate a position of the turntable motor **44**.

[0117] The monitor component **425** can, using the set of data, determine a position of a first component on the electric boom **10** with respect to a second component on the electric boom **10**. For example, the monitor component **425** can determine a position of the turntable **70** (e.g., the first component) with respect to the base assembly **12** (e.g., the second component). For example, the monitor component **425** can determine that the turntable **70** is placed 90 degrees (e.g., the position of the first component) from the base assembly **12**. The position of the first component with respect to the second component can be the same and/or different. For example, the position of the first component can be the same as a position of the second component (e.g., the first component and the second component are aligned). As another example, the position of the first component with respect to the second component can be based on an axis that runs through the first component and an axis that runs through the second component (e.g., an angular difference between the axis of the first component with respect to the axis of the second component). The position of the first component with respect to the second component can be used to determine a quadrant and/or sector (on a coordinate grid) that the first component and/or the second component are located in. For example, the monitor component **425** can determine that the first component is located in the first quadrant and that the second component is located in the third quadrant.

[0118] The monitor component **425** can also determine, using the set of data pertaining to the operation of the electric boom **10**, a first orientation of a first portion of the first component and a first orientation of a second portion of the first component. For example, the monitor component **425** can determine that the portion of the turntable **70** coupled with the platform assembly **16** (e.g., the first portion) has a first orientation and that the portion (e.g., the second portion) of the turntable

70 near the rear of the electric boom **10** has a second orientation. The first orientation and/or the second orientation can be and/or include at least one of a placement of the portion in space (e.g., an area that is occupied by the portion and the location of said area with respect to the electric boom **10**), an elevation (e.g., the platform assembly **16** is raised with respect to base assembly **12**), a configuration of the portion (e.g., how the portion is situated with respect to other components of the electric boom **10**).

[0119] The monitor component **425** can also determine, using the set of data pertaining to the operation of the electric boom **10**, a first orientation of a first portion of the second component and a first orientation of a second portion of the second component. The first orientation and the second orientation of the second component can be and/or include similar characteristics to that of the first orientation and the second orientation of the first component. For example, the first orientation of the second component can indicate that the chassis (e.g., the frame **54**) is offset from the turntable **70**. To continue this example, the first component and the second component can both have a neutral orientation. The neutral orientation can be and/or include at least one of a natural placement (e.g., where components rest while the electric boom **10** is inactive), a placement of the components with respect to a set point (e.g., a central axis of the electric boom **10**), and/or a user-defined placement.

[0120] The neutral orientation of the first component and the neutral orientation of the second component can be used to determine the first orientation and the second orientation of the first component and/or the second component. For example, the orientation of the first component and/or the second component can be considered unchanged when the orientations align with the neutral orientations. Additionally, the orientations of the first component and/or the second component can be relative to their respective neutral orientations.

[0121] The monitor component **425** can, responsive to determining the first orientation and the second orientation for the first component, establish a position of the first component with respect to an axis. The axis can be the central axis described herein and the axis can run, while the first component and the second component are in their respective neutral orientations, through the first portion and the second portion of both the first component and the second component. The position of the first component with respect to the axis can be based on an angular difference between the first component and the axis. For example, the position of the first component with respect to the axis can be determined responsive to determining a difference between a central point on the first component and the axis.

[0122] The monitor component **425** can, using the first orientation of the first portion of the second component and the first orientation of the second portion of the second component, establish a position of the second component with respect to the axis. For example, the monitor component **425** can determine a position of the base assembly **12** with respect to the axis.

[0123] The monitor component **425** can provide, to the video manager **430**, the position of the first component with respect to the second component. For example, the monitor component **425** can provide, to the video manager **430**, a signal that indicates that the first component is the second quadrant and that the second component is in the first quadrant. The video manager **430** can receive, from the monitor component **425**, the position of the first component with respect to the second component.

[0124] The video manager **430** can, using a number of rules, determine a first set of cameras **460** from the cameras **460** that are associated with the position of the first component. The number of rules can be located in memory **420**. The number of rules can be and/or include at least one of logic tables, truth tables, k-maps, Boolean expressions, and/or among other possible rules. For example, the number of rules can be a logic table and the logic table can include which cameras are associated with different sectors. To continue this example, the number of rules can include that if the base assembly **12** is in the first sector and if the turntable **70** is also in the first sector that a first set of cameras **460** of the cameras **460** are associated with that positioning of the first component

(e.g., the turntable **70**) with respect to the second component (e.g., the base assembly **12**).

[0125] The video manager **430** can, responsive to determining which cameras **460** are associated with the position of the first component, retrieve, access and/or otherwise obtain video feeds and/or other data produced by the cameras **460** associated with the position of the first component. The video manager **430** can, responsive to retrieving the data produced by the cameras **460**, communicate with the interface generator **435**. The video manager **430** can provide, to the interface generator **435**, the data produced by the cameras **460**. Additionally, the video manager **430** can provide a signal, to the interface generator **435**, including a data stream that is generated by the cameras **460** (e.g., a live video feed). The video manager **430** can also provide at least one of the position of the first component with respect to the second component, a quadrant including the first component and/or a quadrant including the second component.

[0126] The interface generator **435** can receive, from the video manager **430**, the position of the first component with respect to the second component and the video feed pertaining to the first set of cameras **460**. The interface generator **435** can generate a user interface and/or otherwise generate information that can be used to generate a user interface. For example, the interface generator **435** can provide signals to a display device (e.g., the display device **465**) that causes the display device to display a user interface that includes information generated by the interface generator **435**.

[0127] The interface generator **435** can include the position of the first component with respect to the second component and the video feed of the first set of cameras **460** in the signals that are provided to the display device. The interface generator **435** can, responsive to generating the user interface, communicate with the network interface **445**. For example, the interface generator **435** can provide, to the network interface **445**, the user interface and/or the signals associated with the user interface. The network interface **445** can, responsive to communicating with the interface generator **435**, provide the user interface and/or the signals corresponding to the user to the display device **465**. The display device **465** can, responsive to receiving the user interface, produce, provide, and/or otherwise display the user interface that was generated by the interface generator **435**.

[0128] The monitor component **425** can, responsive to a second operation of the electric boom **10**, receive a second set of data pertaining to the second operation of the electric boom **10**. The monitor component **425** can receive the second set of data from the sensors **455**. The monitor component **425** can, using the second set of data pertaining to the second operation of the electric boom **10**, determine a function associated with the second operation of the electric boom **10**. The function associated with the second operation of the electric boom **10** can be and/or include at least one of the electric boom **10** moving forward (e.g., driving forward), the electric boom **10** moving backwards (e.g., driving in reverse), the turntable **70** swinging (e.g., rotating), the platform assembly **16** being raised, the platform assembly **16** being lowered, and/or among other possible operations and/or combinations of operations.

[0129] The monitor component **425** can, responsive to determining the function associated with the second operation of the electric boom **10**, communicate with the video manager **430**. The monitor component **425** can provide, to the video manager **430**, the function associated with the second operation of the electric boom **10**. The video manager **430** can, responsive to receiving the function associated with the second operation of the electric boom **10**, determine a second set of cameras **460** from the cameras **460** associated with the function. The video manager **430** can determine the second set of cameras **460** using the number of rules. For example, the number of rules can include logic tables that indicate that when the first component is in the first quadrant, when the second component is in the second quadrant, and when the base assembly **12** is being raised that given cameras **460** of the cameras **460** should be used. The video manager **430** can retrieve, access, and/or otherwise obtain video feeds and/or other data produced by the second set of cameras **460**.

[0130] The video manager **430** can, responsive to determining the second set of cameras **460**, communicate with the interface generator **435**. The video manager **430** can provide, to the interface

generator **435**, the data produced by the second set of cameras **460**. The video manager **430** can also provide the function associated with the second operation of the electric boom **10**.

Additionally, the video manager **430** can provide a signal, to the interface generator **435**, including a data stream that is generated by the second set of cameras **460** (e.g., a live video feed, a field of view, etc.). The video manager **430** can also provide an orientation, a position, a configuration and/or among other possible component information that can change responsive to the second operation of the electric boom **10**.

[0131] The interface generator **435** can receive, from the video manager **430**, the data pertaining to second set of cameras **460**. The interface generator **435** can generate a user interface, update the user interface previously generated by the interface generator **435** and/or otherwise generate information that can be used to generate a user interface. The user interface can replace at least one video feed provided by the first set of cameras **460** with at least one video feed provided by the second set of cameras **460**. The interface generator **435** can provide signals to the display device **465** that causes the display device **465** to display a user interface including the at least one video feed of the second set of cameras **460**.

[0132] The registration manager **440** can add, adjust, delete, modify and/or otherwise configure the cameras **460**. For example, the registration manager **440** can update the number of cameras **460** to include a new camera **460** responsive to the new camera **460** being added to the electric boom **10**. The registration manager **440** can associated the cameras **460** with a position on the electric boom **10**. For example, the registration manager **440** can associate a first camera **460** with being positioned on center portion of a counterweight of the turntable **70**.

[0133] The registration manager **440** can detect, responsive to receiving a request to initiate a registration process from a number of cameras **460**, the number of cameras **460**. For example, the registration manager **440** can detect the request responsive to an operator of the electric boom **10** and/or the user device **485** selecting an icon on a user interface. The selection of the icon can result in the registration manager **440** detecting the number of cameras **460**. For example, the registration manager **440** can scan, explore, search, and/or otherwise survey the network **480** to detect the number of cameras **460**. Additionally, the registration manager **440** can detect the number of cameras **460** responsive to the number of cameras being coupled with the electric boom **10** via the Ethernet switch **475**.

[0134] The registration manager **440** can determine, using information pertaining to the number of cameras **460**, that a camera **460** of the number of cameras **460** was previously absent from the network **480**. For example, the registration manager **440** can include a database of all configured cameras **460** that connected to the network **480** and the registration manager **440** can determine the camera **460** is not in the database. To continue this example, the registration manager **440** can determine that the camera **460** was previously absent from the network.

[0135] The registration manager **440** can, responsive to determining that the camera **460** was previously absent from the network, generate an identifier for the camera **460** for use in the network **480**. The identifier can be and/or include at least one of an IP address, a numeric string, an alphanumeric string and/or among other possible strings of data that can uniquely identify the camera **460** within the network **480**. The registration manager **440** can, responsive to generating the identifier for the camera **460**, retrieve a video feed pertaining to the camera **460**. The registration manager **440** can provide the video feed to the interface generator **435**.

[0136] The interface generator **435** can, responsive to receiving the video feed, generate a user interface including the video feed. The user interface can also include additional information pertaining to the camera **460**. For example, the user interface can include selectable portions that can be used to indicate placement of the camera **460** on the electric boom **10**. The interface generator **435** can provide signals to the display device **465** that causes the display device **465** to display the user interface including the video feed of the camera **460**.

[0137] The registration manager **440** can detect, responsive to an interaction with the user interface,

an indication identifying the position of the camera **460**. For example, the registration manager **440** can detect an operator of the display device **465** selecting an icon that pertains to the location of the camera **460**. The icon could display a certain position on the electric boom **10** and/or other receive a user input identifying the certain on the electric boom **10**. The registration manager **440** can, responsive to identifying the position of the camera **460**, associate with the camera **460** with at least one component of the electric boom **10**. For example, the indication can include that the camera **460** is disposed on the platform assembly **16**.

[0138] The registration manager **440** can, responsive to associating the camera **460** with the at least one component of the electric boom **10**, update a record to reflect that the camera **460** is associated with the at least one component of the electric boom **10**. For example, the registration manager **440** can update a data file that includes information pertaining to the cameras **460** to include information pertaining to the new camera **460**. The registration manager **440** can, using the record, identify a second camera **460** that is associated with the at least one component of the electric boom **10**. For example, the record can indicate the second camera **460** is disposed on the turntable **70** and the camera **460** can also be disposed on the turntable **70**.

[0139] The registration manager **440** can, responsive to identifying the second camera **460** that is also associated with the at least one component of the electric boom **10**, retrieve data and/or a video feed generated by the camera **460** and the second camera **460**. The registration manager **440** can provide the video feed to the interface generator **435**. The interface generator **435** can, responsive to receiving the video feed, generate a user interface including the video feed of the camera **460** and the second camera **460**. The registration manager **440** can determine, responsive to an interaction with the user interface, a relative position of the camera **460** with respect to the second camera **460**. For example, the interaction can specify that the camera **460** is position on the rear of the base assembly **12** and the second camera **460** is positioned on the front of the base assembly **12**.

[0140] FIG. 5 depicts a block diagram of a system **500** including components of the electric boom **10**, according to an exemplary embodiment. The system **500** can include components that are connected via a network (e.g., the network **480**). The lines connecting the components to one another can be wired connections (e.g., an Ethernet cable that is electrically coupling the components to one another) and/or a wireless connections. The system **500** can include the electric boom **10**. The electric boom **10** can include the platform assembly **16**, the turntable **70**, the chassis **54** (e.g., the frame **54**), and at least one video processing unit (VPU) **505**. While FIG. 5 depicts the VPU **505** as being separate from the additional components of the electric boom **10**, the VPU **505** can included in at least one component of the electric boom **10**. For example, the VPU **505** can be included in the display device **465**. The VPU **505** can be and/or include at least one a graphical processing unit (GPU), a tensor processing unit (TPU), a neural processing unit (NPU), and/or among other possible processing devices and/or components. The VPU **505** can perform object detection. For example, the VPU **505** can detect that an object is near the electric boom **10**.

[0141] The display device **465** and the cameras **460** of the platform assembly **16** are shown to be connected with the additional components of the electric boom via the Ethernet switch **475**. Similarly, the cameras **460** of the turntable **70** are shown to also be connected with the additional components of the electric boom via the Ethernet switch **475**. Additionally, the chassis **54** is shown to include a rotary coupler **510** that is connected to the additional components of the electric boom **10** and that the cameras **460** of the chassis **54** are connected with the rotary coupler **510** via the Ethernet switch **475**.

[0142] FIG. 6 is an aerial view of the electric boom **10**, according to an exemplary embodiment. FIG. 6 depicts an example of the turntable **70** having been rotated, and an example of the turntable **70** in a position relative to the base assembly **12**. FIG. 6 depicts a counterweight **605** of the turntable **70** including cameras **460** (e.g., CAM 1, CAM 2, and CAM 3), the base assembly **12** including cameras **460** (e.g., CAM 4, and CAM 5), and the platform assembly **16** including cameras **460** (e.g., CAM 6). While FIG. 6 depicts the components including a certain number of

cameras **460** the components can include any possible number of cameras **460**.

[0143] The placement of the cameras **460**, on their respective components of the electric boom **10**, provide different fields of view and/or otherwise produce video feeds that captures different areas around the electric boom **10** based on the orientation of the components of the electric boom **10**. For example, CAM **1**, as illustrated in FIG. **6**, can produce a video feed that can capture a different area around the electric boom **10** with respect to a video feed produced by CAM **5**. As another example, CAM **1** can produce a field of view that can capture a difference area around the electric boom **10** with respect to a field of view produced by the CAM **5**.

[0144] As described herein, the video manager **430** can use a number of rules to determine video feeds that can be used and/or included in user interfaces. Table 1 provides an example of the number of rules being a logic table. Table 1 is produced below:

TABLE-US-00001 TABLE 1 Quadrant Logic Table A B Quadrant 0 0 1 0 1 2 1 0 3 1 1 4

[0145] Table 1 depicts an example logic table that can be used to determine a position, with respect to a quadrant, of a component of the electric boom **10**. For example, the turntable **70** can include a sensor and the sensor can produce a two bit signal that can be used to identify the quadrant of the turntable **70**. For example, as the turntable **70** rotates the two bit signal can be incremented and/or otherwise changed to reflect the rotation of the turntable **70**. Table 1 shows that the component of the electric boom **10** can be in quadrant 1 when the two bit signal equals 0, that the component of the electric boom **10** can be in quadrant 2 when the two bit signal equals 1, component of the electric boom **10** can be in quadrant 3 when the two bit signal equals 2, and that the component of the electric boom **10** can be in quadrant 4 when the two bit signal equals 3. The logic depicted in table 1 can be used by the control system **403** to perform any of the functionality described herein.

[0146] Table 2 depicts another logic table that can be used to determine the position of the first component and the second component of the electric boom **10**. While tables 2 includes the components as the turntable and the base assembly, the components used in the table 2 can be any possible components of the electric boom **10**. Table 2 is produced below:

TABLE-US-00002 TABLE 2 Base Base Camera Assembly Turntable Turntable Assembly Zone
Logic Logic Quadrant Quadrant Activated 0 0 0 0 1 1 0 0 0 0 1 2 1 1 0 0 1 0 3 1 2 0 0 1 1 4 1 3 0 1
0 0 1 2 4 0 1 0 1 2 2 5 0 1 1 0 3 2 6 0 1 1 1 4 2 7 1 0 0 0 1 3 8 1 0 0 1 2 3 9 1 0 1 0 3 3 10 1 0 1 1 4
3 11 1 1 0 0 1 4 12 1 1 0 1 2 4 13 1 1 1 0 3 4 14 1 1 1 1 4 4 15

[0147] Table 2 depicts the logic table including logic pertaining to the turntable **70**, the base assembly **12**, and that the logic can be used to determine a quadrant for the turntable **70**, determine a quadrant for the base assembly **12**, and determine a camera zone. The logic depicted in table 2 can be used by the control system **403** to perform any of the functionality described herein.

[0148] Table 3 depicts an example of the number of rules identifying which cameras correspond to which zones. Table 3 is produced below:

TABLE-US-00003 TABLE 3 Camera Zone Cameras in Zone 0 1, 2, 3, 6 1 1, 4, 5, 6 2 1, 2, 3, 6 3 1, 2, 3, 4, 6 4 1, 2, 3, 4, 5, 6 5 1, 2, 3, 6 6 1, 2, 3, 5, 6 7 1, 2, 4, 5, 6 8 1, 3, 5, 6 9 1, 2, 3, 4, 5, 6 10 1, 2, 3, 6 11 1, 2, 3, 4, 6 12 1, 2, 3, 4, 5, 6 13 1, 3, 5, 6 14 1, 2, 3, 4, 5, 6 15 1, 2, 3, 6

[0149] The video manager **430** can use the information included in the table 3 to determine which cameras **460** of the cameras **460** pertain to different camera zones. As shown above in table 2, the camera zones can be determined using the information in table 2 and once the camera zones are determined, the corresponding cameras can be identified. While table 3 includes given zones including given cameras, the camera zones can include any possible combinations of cameras including some zones having extra cameras and/or some zones having less cameras in relation to the number of cameras depicted in a given zone of table 3.

[0150] FIG. **7** is an aerial view of the electric boom **10**, according to an exemplary embodiment. As described herein, the electric boom **10** and/or the components thereof can have and/or include a neutral orientation. FIG. **7** depicts an example of the turntable **70** and the base assembly **12** in their respective neutral orientations. FIG. **7** also depicts an axis **725** that runs through the center of the

turntable **70** and the base assembly **12**. The axis **725** can be the central axis described herein. FIG. **7** also depicts an example of the electric boom **10** located on a coordinate grid and/or system include quadrants 1, 2, 3, 4. The quadrants can be the quadrants and/or sectors described herein. Lines **710** and **720** define quadrant 1, lines **720** and **715** define quadrant 2, lines **715** and **705** define quadrant 3, and lines **705** and **710** define quadrant 4. While FIG. **7** depicts quadrants 1 and 3 as being larger than quadrants 2 and 4, the relative size of one quadrant with respect to another quadrant can be and/or include any possible combination. For example, the 4 quadrants can be the same size.

[0151] FIG. **7** also depicts an example of the turntable **70** and the base assembly being located in the first quadrant. However, any possible configuration and/or orientation of the coordinate grid can be possible. For example, the location of the base assembly **12** and the turntable **70**, as described above, is based on the position of a frontward portion of the turntable **70** and/or the base assembly **12**. The location of the base assembly **12** and/or the turntable **70** can be based on the position of any portion of the turntable **70** and/or the base assembly **12**. For example, a rearward portion of the turntable **70** can be used to determine the position of the turntable **70** with respect to the coordinate grid.

[0152] FIG. **8** is an aerial view of the electric boom **10**, according to an exemplary embodiment. FIG. **8** depicts an example of the turntable **70** having been rotated from its neutral orientation and an example of the base assembly **12** in its neutral configuration. As described herein the positioning of the turntable **70** (e.g., the first component) with respect to the base assembly **12** (e.g., the second component) can determine which video feeds that are produced by the cameras **460** are displayed in a user interface. For example, the layout of the electric boom **10** and/or the components of thereof, as illustrated in FIG. **8**, can be used by the control system **405**.

[0153] FIG. **8** depicts an example of the turntable **70** having been rotated from quadrant 1 to quadrant 2. The turntable **70** being positioned in quadrant 2 and the base assembly **12** being positioned in quadrant 1 can be used by the control system **405** to determine which video feeds to display in a user interface. For example, the axis **725** is shown to be running through the base assembly **12** and an axis **805** is shown to be running through the turntable **70**. An angle between the axis **725** and the axis **805** can be used to determine the position (e.g., which quadrant) of the turntable **70** relative to the base assembly **12**.

[0154] FIG. **9** depicts a flow chart illustrating a method **900** of using control logic to determine video feeds, according to an exemplary embodiment. The method **900** can be performed by the control system **403** and/or a component thereof. For example, the video manager **430** can perform at least one step of the method **900**. The method **900** can begin at step **905**. Step **905** can include receiving a vehicle input. The vehicle input can be at least one of an operator of the electric boom **10** interacting with the I/O device **470**, an operator of the user device **485** selecting an icon included on a user interface, and/or an operator of the display device **465** selecting an icon included on a user interface. For example, the vehicle input can be an operator interacting with a joystick of the electric boom **10** (e.g., the I/O device **470**). The interaction can be the operator using the joystick to move, control, and/or otherwise perform an operation using the electric boom **10**.

[0155] The method **900** can continue to step **910**. Step **910** can be a decision step (e.g., the video manager **430** can make a decision and the decision can dictate the next step in the method **900**). The video manager **430** can, at step **910**, determine whether an option has been selected for the control system **403** to produce automatic video feeds and/or whether an operator of the electric boom **10** has selected an option to manual select the video feed. The method **900** can move on to step **915** responsive to the video manager **430** determining that the option to produce automatic video feeds has been selected. The method **900** can move to step **920** responsive to the video manager determining that the option to manual select the video feed has been step.

[0156] Step **915** can include the video manager **430** receiving information pertaining to the position of the electric boom **10** and/or a component thereof (e.g., the first component and the second

component, receiving information pertaining to a function that was performed responsive to the vehicle input received at step **905**, and/or any other possible information described herein. The video manager **430** can use the number of rules to determine which quadrant the components of the electric boom **10** are in. The video manager **430** can, responsive to determining the quadrants, determine the cameras **460** to retrieve video feeds from. Step **915** can also include the video manager **430** sending control signals to the interface generator **435** and the interface generator **435** can send signals to the display device **465** causing the display device **465** to display a user interface including the video feeds produced by the cameras **460** that were determined at step **915**.

[0157] Step **920** can include the video manager **430** communicating with the interface generator **435** and the interface generator **435** sending signals to the display device **465** causing the display device **465** to display a user interface including options for the operator to interact with to then manual select cameras.

[0158] FIG. **10** is a block diagram of a system **1000** including components of the electric boom **10**, according to an exemplary embodiment. FIG. **10** depicts an example of how the display device **465** can be connected and/or otherwise in communication with the cameras **460**. FIG. **10** shows that the display device **465** is connected with the Ethernet switch **475**. The display device being connected with the Ethernet switch **475** can result in the display device **465** being included in the network **480**.

[0159] The Ethernet switch **475** can include at least one port **1005**. The ports **1005** can connect components of the electric boom **10** with the Ethernet switch **475** and then also connect the components with the network **480**. FIG. **10** depicts Cam **1**, Cam **2**, and Cam **3** (e.g., a number of cameras **460**) connected with, via the ports **1005**, the Ethernet switch **475**. The Ethernet switch **475** can connect the display device **465** with the number of cameras **460**. The display device **465** and the number of cameras **460** can include individual IP addresses and a subnet mask. The individual IP addresses can be the IP addresses generated during the registration process described herein.

[0160] FIG. **11** is a flow chart of a method **1100** of registering cameras with a network, according to an exemplary embodiment. The method **1100** can be and/or include the registration process described herein. The control system **403** and/or a component thereof can perform at least one step included in the method **1100**. For example, the registration manager **440** can perform at least one step of the method **1100**.

[0161] The method **1100** can begin at step **1105**. Step **1105** can include initiated a camera location registration method (CLRM) sequence. The CLRM sequence can be initiated responsive to an operator of the electric boom **10** and/or the user device **485** selecting an icon and/or an option displayed on a user interface. The operator can select the option responsive to a camera **460** being added to the electric boom **10**. For example, the camera **460** can be disposed on the electric boom **10** and prior to using the camera **460** the can **460** can be configured and/or otherwise registered.

[0162] The method **1100** can continue to step **1110**. Step **1110** can include searching for cameras **460** that included and/or otherwise connected to the network **480**. The method **1100** can continue to step **1115**, responsive to searching for the cameras **460**. Step **1115** can include collecting and counting the MAC addresses from the cameras **460**. The method **1100** can continue to step **1120**, responsive to the accumulating the information pertaining to the cameras **460**.

[0163] Step **1120** can be a decision step (e.g., the registration manager **440** can make a decision). Step **1120** can include the registration manager **440** determining whether any of the MAC addresses collected in step **1115** are new (e.g., a camera **460** was previously absent from the network **480**). The method **1100** can continue to step **1125** responsive to the registration manager **440** determining that at least one MAC address is new. The method **1100** can continue to step **1135** responsive to the registration manager **440** determining that none of the MAC addresses are new.

[0164] Step **1125** can include the registration manager **440** assign an IP address to the camera **460** corresponding to the MAC address that was determined to be new in step **1120**. Step **1125** can be repeated for each additional camera **460** that included a MAC address that was determined to be

new in step **1120**. The method **1100** can continue to step **1130**.

[0165] Step **1130** can include the registration manager **440** calibrating the cameras **460** that had IP addresses assigned to them in step **1125**. The calibration can be and/or include downloading software, running a diagnostic test, and/or any possible actions that can be taken to link the cameras **460** with the network **480**. The method **1100** can continue to step **1135**.

[0166] Step **1135** can be a decision step (e.g., the registration manager **440** can make a decision). Step **1135** can include the registration manager **440** determining whether the number of devices (e.g., cameras) not mapped is equal to 0. As mentioned above, step **1135** can be reached after step **1120** and/or after step **1130**. The registration manager **440** uses step **1135** to then determine which cameras **460** to map. The method **1100** can continue to step **1140** responsive to the registration manager **440** determining that the number of devices not mapped is not equal to 0 (e.g., at least one camera **460** has not been mapped yet). The method **1100** can continue to step **1155** responsive to the registration manager **440** determining that the number of devices not mapped is equal to 0.

[0167] Step **1140** can include the registration manager **440** sorting the cameras **460**, that have not been mapped yet, by IP address and the interface generator **435** can cause the display device **465** to display a user interface including the video feeds for at least one of the cameras **460**. The method **1100** can continue to step **1145**.

[0168] Step **1145** can include mapping the camera locations. For example, the operator of the electric boom **10** can select icons displayed on the user interface to identify locations pertaining to the cameras (e.g., the cameras can be mapped). The locations pertaining to the cameras can be and/or include where the cameras are disposed and/or otherwise placed on the electric boom **10**. The method **1100** can continue to step **1150**.

[0169] Step **1150** can include displaying the video feeds of the cameras in a sort order. The sorted order can be based on the sorted order of the IP addresses that was performed in step **1140**. Step **1155** can end and/or otherwise conclude the CLRM sequence and/or the registration process described herein.

[0170] FIG. **12** depicts a user interface **1200**, according to an exemplary embodiment. The user interface **1200** can be provided, produced, generated and/or otherwise displayed by the display device **465** and/or the user device **485**. The user interface **1200** can be generated by the interface generator **435** and the interface generator **435** can provide signals to the display device **465** and/or the user device **485** causing the display device **465** and/or the user device **485** to display the user interface **1200**.

[0171] The user interface **1200** can include at least one icon (e.g., icons **1205**, **1210**, **1215**, **1220**, **1225**, **1240**, **1245**, **1250**, and **1255**), and at least one video feed (e.g., video feeds **1230** and **1235**). The operator of the device displaying the user interface **1200** can select, interact with and/or otherwise engage with at least one of the icons. The icon **1205** can provide a visual representation of the orientation of the chassis **54**. The icon **1205** can be generated using information determined by the control system **403**. For example, the monitor component **425** can determine the orientation of the chassis **54** and the monitor component **425** can provide an indication of the orientation of the chassis **54** to the interface generator **435**. The icon **1205** depicts an example of the chassis **54** in an offset and/or rotated orientation.

[0172] The icons **1210**, **1215**, and **1220** can represent and/or otherwise identify given cameras **460** that are providing the video feeds **1230** and **1235**. The icons **1215** and **1220** are shown to be a different color than that of icon **1210**. The color difference can indicate which cameras **460** are producing the video feeds **1230** and **1235**. The icon **1225** can be selected to reach a home page and/or among other possible pages and/or user interfaces. The icon **1240** can be selected to open and/or otherwise produce a drop down menu and/or a window. The icons **1245** and **1250** can be selected to produce and/or otherwise provide a list of options that can be performed by the highlighted component of the electric boom **10**. FIG. **12** shows the icon **1245** to be associated with the chassis **54** of the electric boom and shows the icon **1250** to be associated with the turntable **70**,

the lift assembly **14**, and/or the platform assembly **16**. The icon **1255** can be selected to open and/or otherwise produce a settings window. For example, the icon **1255** can be selected and an option to turn on manual selection of video feeds can then be displayed on the user interface **1200**. The video feeds **1230** and **1235** are shown to be producing images captured by the cameras **460** indicated by icons **1215** and **1220**. The video feeds **1230** and **1235** can be and/or include the video feeds described herein. The user interface **1200** can also include, produce, provide, and/or otherwise display any information described herein.

[0173] FIG. **13** depicts a user interface **1300**, according to an exemplary embodiment. The user interface **1300** can be provided, produced, generated and/or otherwise displayed by the display device **465** and/or the user device **485**. The user interface **1300** can be generated by the interface generator **435** and the interface generator **435** can provide signals to the display device **465** and/or the user device **485** causing the display device **465** and/or the user device **485** to display the user interface **1300**. The user interface **1300** can be an updated user interface with respect to the user interface **1200** (e.g., the user interface **1200** was updated to display the information shown in FIG. **13**). The user interface **1300** can also be generated responsive to an operation of the electric boom **10**.

[0174] The user interface **1300** can include at least one icon **1305**, at least one video feed **1330**, and at least one video feed **1335**. The icon **1305** can provide a visual representation of the orientation of the chassis **54**. The icon **1305** can be generated using information determined by the control system **403**. For example, the monitor component **425** can determine the orientation of the chassis **54** and the monitor component **425** can provide an indication of the orientation of the chassis **54** to the interface generator **435**. The icon **1305** can also be an updated version of the icon **1205** and the icon **1305** can be generated responsive to the orientation of the chassis **54** changing (e.g., the chassis **54** was moved). The icon **1305** depicts an example of the chassis **54** in an inline orientation. The video feeds **1330** and **1335** can include images captured by the cameras **460** associated with the orientation of the chassis **54**.

[0175] FIG. **14** depicts a user interface **1400**, according to an exemplary embodiment. The user interface **1400** can be provided, produced, generated and/or otherwise displayed by the display device **465** and/or the user device **485**. The user interface **1400** can be generated by the interface generator **435** and the interface generator **435** can provide signals to the display device **465** and/or the user device **485** causing the display device **465** and/or the user device **485** to display the user interface **1400**. The user interface **1400** can be an updated user interface with respect to the user interface **1200** (e.g., the user interface **1200** was updated to display the information shown in FIG. **14**) and/or the user interface **1300** (e.g., the user interface **1300** was updated to display the information shown in FIG. **14**). The user interface **1400** can also be generated responsive to an operation of the electric boom **10**.

[0176] The user interface **1400** can include at least one icon **1405**, at least one video feed **1430**, and at least one video feed **1435**. The icon **1405** can provide a visual representation of the orientation of the chassis **54**. The icon **1405** can be generated using information determined by the control system **403**. For example, the monitor component **425** can determine the orientation of the chassis **54** and the monitor component **425** can provide an indication of the orientation of the chassis **54** to the interface generator **435**. The icon **1405** can also be an updated version of the icon **1205** and/or icon **1305**. The icon **1405** can be generated responsive to the orientation of the chassis **54** changing (e.g., the chassis **54** was moved). The icon **1405** depicts an example of the chassis **54** in an orthogonal orientation. The video feeds **1430** and **1435** can include images captured by the cameras **460** associated with the orientation of the chassis **54**.

[0177] FIG. **15** depicts a user interface **1500**, according to an exemplary embodiment. The user interface **1500** can be provided, produced, generated and/or otherwise displayed by the display device **465** and/or the user device **485**. The user interface **1500** can be generated by the interface generator **435** and the interface generator **435** can provide signals to the display device **465** and/or

the user device **485** causing the display device **465** and/or the user device **485** to display the user interface **1500**. The user interface **1500** can be the user interface described herein with respect to the CLRM sequence of the method **1100** and/or the registration process described herein.

[0178] The user interface **1500** can a plurality icons (e.g., icons **1505**, **1510**, **1515**, **1520**, **1525**, **1530**, **1540**, **1545**, **1550**, **1555**, **1560**, **1565**, and **1570**). The operator of the device displaying the user interface **1500** can select, interact with and/or otherwise engage with at least one of the icons. The icon **1505** can provide and/or otherwise depict a visual representation of the chassis **54**. The icon **1505** can also depict a visual representation of any other element and/or component included in the electric boom **10**. The icon **1505** is shown to be associated with icons **1510** and **1515**. The icons **1510** and **1515** are associated with cameras **460** that are disposed on respective portions of the chassis **54**. The icon **1510** indicates a camera **460** that is disposed on a frontward portion of the chassis **54** and the icon **1515** indicates a camera **460** that is disposed on a rearward portion of the chassis **54**.

[0179] The icon **1520** can provide and/or otherwise depict a visual representation of the turntable **70**. The icon **1520** can also depict a visual representation of any other element and/or component included in the electric boom **10**. The icon **1520** is shown to be associated with icons **1525**, **1530**, and **1535**. The icons **1525**, **1530**, and **1535** are associated with cameras **460** that are disposed on respective portions of counterweight **605** of the turntable **70**. The icon **1525** indicates a camera **460** that is disposed on a right portion of the counterweight **605**. The icon **1530** indicates a camera **460** that is disposed on a center portion of the counterweight **605**. The icon **1535** indicates a camera **460** that is disposed on a left portion of the counterweight **605**. The icon **1535** is shown to be highlighted and/or otherwise selected. The icon **1535** being selected indicates that the video feed shown in FIG. **15** is associated with the camera **460** that is indicated by the icon **1530**.

[0180] The icon **1540** can be selected to reach a home page and/or among other possible pages and/or user interfaces. The icon **1545** can provide an indication that the electric boom **10** is a calibration mode. The icon **1550** can be selected to open and/or otherwise produce a drop down menu and/or a window. The icon **1555** can be selected to switch the video feed to a previously shown video feed. The icon **1560** can be selected to set and/or otherwise map the camera **460** whose video feed is displayed in the user interface **1500** with the icon that indicates said camera. The icon **1565** can be selected to switch the video feed to a video that is yet to have been shown. The icon **1570** can be selected to reset the calibration process.

[0181] FIG. **16** depicts a perspective view of the electric boom **10**. The perspective view of the electric boom **10**, as shown in FIG. **16**, includes a transparent view of given portions of the electric boom **10**. A number of cables **1605** can be disposed throughout the electric boom **10**. The cables **1605** can be and/or include at least one of Ethernet cables, fiber optic cables, Coaxial cables, twisted pair (shielded and/or unshielded) cables, and/or among other possible cables that can be used in networks. The cables **1605** can be electrically coupled with at least one of the sensors **455**, the cameras **460**, the display device **465** and/or the Ethernet switch **475**. FIG. **16** depicts an example of the cables **1605** electrically coupled with the cameras **460** and the display device **465**. The cables **1605** can electrically couple the cameras **460** and/or the display device **465** with the Ethernet switch **475**. The electric coupling, via the cables **1605**, of the cameras **460** and the display device **465** can connect the cameras **460** and the display device **465** with the network **480**.

[0182] FIG. **17** depicts a perspective view of the electric boom **10**, according to an exemplary embodiment. The perspective view of the electric boom **10**, as shown in FIG. **17**, includes a transparent view of given portions of the electric boom **10**. The cables **1605** can be electrically coupled with the cameras **460**, the Ethernet switch **475** and/or the rotary coupler **510**. The cameras **460** can be disposed on the chassis **54**. The cables **1605** can electrically couple the cameras **460** with the Ethernet switch **475** and the cables **1605** can electrically couple the Ethernet switch **475** with the rotary coupler **510**.

[0183] FIG. **18** is a perspective view of the platform assembly **16**. The platform assembly **16** can

include the display device **465**. The display device **465** can be electrically coupled with the cables **1605**. The cables **1605** can electrically couple the display device **465** with the Ethernet switch **475**. [0184] FIG. **19** is a perspective view of the electric boom **10**, according to an exemplary embodiment. The cables **1605** can be electrically coupled with the Ethernet switch **475**. The Ethernet switch **475** can be disposed within the counterweight **605**. The counterweight **605** can be included in the turntable **70**. The cameras **460** can be disposed with the counterweight **605** and the cameras **460** can produce video feeds pertaining to an area around the turntable **70**. The cameras **460** can be electrically coupled with the cables **1605** and the cables **1605** can electrically couple the cameras **460** with the Ethernet switch **475**.

[0185] FIG. **20** is a perspective view of the electric boom **10**, according to an exemplary embodiment. The electric boom **10** can include at least one moveable elements (e.g., the turntable **70**, the chassis **54**, the base assembly **12**, the platform assembly **16**, etc.) The moveable elements can have at least portion. For example, the moveable elements can have a first portion and a second portion. The cameras **460** can be disposed on the moveable elements. For example, the cameras can be disposed on the second portion of the moveable elements. The cameras **460** can produce a number of video zones (e.g., video zones **2005**, **2010**, **2015**, **2020**, **2025**, and **2030**). The video zones can capture an area around the electric boom **10**. The video zones can include at least one field of view. The field of views can extend away from the electric boom **10**. The area around the electric boom **10** can be and/or include a geographic area including the electric boom **10**, an environment including the electric boom **10**, and/or any other possible space that can include and/or otherwise be occupied by the electric boom **10**.

[0186] The video zone **2005** can be a video zone captured by a first camera **460**. The first camera **460** can be disposed on a frontward portion (e.g., a first portion) of the platform assembly **16**. The video zone **2005** can capture an area that is in front of the frontward portion of the platform assembly **16**. The video zone **2005** can be and/or include a field of view (FOV) with respect to the platform assembly **16**.

[0187] The video zone **2010** can be a video zone captured by a second camera **460**. The second camera **460** can be disposed on a frontward portion (e.g., a first portion) of the chassis **54**. The video zone **2010** can capture an area that is in front of the frontward portion of the chassis **54**. The video zone **2020** can be a video zone captured by a third camera **460**. The third camera **460** can be disposed on a rearward portion (e.g., a second portion) of the chassis **54**. The video zone **2020** can capture an area that is behind the rearward portion of the chassis **54**. The video zone **2010** and the video zone **2020** can be and/or include a FOV with respect to the chassis **54**.

[0188] The video zone **2015** can be a video zone captured by a fourth camera **460**. The fourth camera **460** can be disposed on a first rearward lateral portion (e.g., a first portion) of the turntable **70**. The video zone **2015** can capture an area that is behind the first rearward lateral portion of the turntable **70**. The video zone **2025** can be a video zone captured by a fifth camera **460**. The fifth camera **460** can be disposed on a second rearward lateral portion of the turntable. The video zone **2025** can capture an area that is behind the second rearward lateral portion of the turntable **70**. The video zone **2030** can be a video zone captured by a fifth camera **460**. The fifth camera **460** can be disposed on a center rearward portion of the turntable **70**. The video zone **2030** can capture an area that is behind the center rearward portion of the turntable **70**.

[0189] The video zone **2015**, **2025**, and **2030** can be and/or include a FOV with respect to the turntable **70**. The FOV with respect to the base assembly **12**, the FOV with respect to the chassis **54**, and the FOV with respect to the turntable **70** can be and/or include a FOV with respect to the electric boom **10**. At least one video zone can change responsive to a first moveable element moving relative to a second moveable element. For example, the turntable **70** (e.g., the first moveable element) can rotate relative to the chassis **54** (e.g., the second moveable element) and the turntable **70** rotating can result in the cameras **460** disposed on the turntable **70** no longer capturing the areas depicted in FIG. **11** (e.g., the turntable **70** is now occupying a different space).

[0190] FIG. 21 is a perspective view of the electric boom 10, according to an exemplary embodiment. The cameras 460 can be disposed on a portion of the turntable 70. For example, the cameras 460 can be disposed on a rearward portion of the turntable 70. The cameras 460 can produce video zones and the video zones can capture an area that is proximate to the turntable 70 (e.g., the area in view of the cameras 460). The video zones produced by the cameras 460 can be combined to create a FOV 2105. The FOV 2105 can be and/or include areas that are captured by the cameras 460.

[0191] FIG. 22 is a perspective view of the electric boom 10, according to an exemplary embodiment. The sensors 455 can be disposed on the electric boom 10. For example, the sensors 455 can be disposed on the turntable 70 (e.g., a first moveable element) and the chassis 54 (e.g., a second moveable element). The sensors can produce at least one proximity zone 2205. The proximity zones 2205 can encompass an area around the electric boom 10. For example, a first proximity zone 2205 can include a circle with a diameter surrounding the electric boom 10.

[0192] The detection of an object, within a given proximity zone 2205, can be used to determine a distance between the object and the electric boom 10. For example, the first proximity zone 2205 can have a radius starting at the electric boom 10 and ending at half the diameter of the proximity zone. When an object is detected at the first proximity zone 2205 the distance of the object can be similar to the radius of the proximity zone 2205. The area captured by the video zone 2030 can include a portion of the proximity zones 2205.

[0193] FIG. 23 is a perspective of view the electric boom 10, according to an exemplary embodiment. The cameras 460 disposed on the chassis 54 can produce the video zones 2010 and the video zones 2020. The video zone 2010 can capture a frontward area of the electric boom 10 and the video zone 2020 can capture a rearward area of the electric boom 10.

[0194] FIG. 24 is a perspective view of the electric boom 10, according to an exemplary embodiment. The sensors 455 can be disposed on the chassis 54. The sensors 455 can produce the proximity zones 2205. The proximity zones 2205, the video zone 2010 and the video zone 2020 can encompass and capture the area of the electric boom 10 with respect to the chassis 54.

[0195] FIG. 25 is a perspective view of a panel 2505, according to an exemplary embodiment. The turntable 70 can include the panel 2505. The panel 2505 can be and/or include the counterweight 605. The panel 2505 can include at least one assembly 2510. The assembly 2510 can be and/or include at least one of a body, a housing, a structure and/or a frame. The assembly 2510 can include at least one camera 460. The assemblies 2510 can be located at various portions of the panel 2505. For example, a first assembly 2510 can be disposed on a rearward lateral portion of the panel 2505 and a second assembly 2510 can be disposed on a center rearward portion of the panel 2505.

[0196] FIG. 26 is a perspective view of the panel 2505, according to an exemplary embodiment. The assembly 2510 can include the cameras 460. The location of the cameras 460, on the assembly 2510, can dictate the video zones produced by the cameras 460. FIG. 26 depicts an example of the cameras 460 disposed towards the center of the assembly 2510. The cameras being disposed towards the center of the assembly 2510 can result in the video feeds produced by the cameras 460 capturing a rearward view of the turntable 70.

[0197] FIG. 27 is a perspective view of the panel 2505, according to an exemplary embodiment. The assembly 2510 can include the cameras 460 and the sensors 455. The sensors 455 and the cameras 460 can be electrically coupled with the cables 1605. The cables 1605 can electrically couple the sensors 455 and the cameras 460 with the Ethernet switch 475.

[0198] FIG. 28 is a perspective view of the panel 2505, according to an exemplary embodiment. The assembly 2510 can include the cameras 460 and the sensors 455. The sensors 455 and the cameras 460 can be electrically coupled with the cables 1605. The placement of the sensors 455 and/or the placement of the cameras 460 in the assembly 2510 can dictate the proximity zones produced by the sensors 455 and the video zones produced by the cameras 460.

[0199] FIG. 29 is a perspective view of the assembly 2510, according to an exemplary

embodiment. The assembly **2510** can include at least one aperture **2905**. The apertures **2905** can provide a void and/or an opening for at least one of the cameras **460** and/or the sensors **455**.

[0200] FIG. **30** is a perspective view of the electric boom **10**, according to an exemplary embodiment. The sensors **455** and the cameras **460** can produce at least one zone **3005**. The zones **3005** can be and/or include the proximity zones and/or the video zones. As the moveable elements of the electric boom **10** move the zones **3005** can change and/or be maintained. For example, a proximity zone produce by the sensors **455** disposed on the lift assembly **14** can change responsive to the lift assembly **14** being raised and/or lowered. As another example, the video zones produced by the cameras **460** disposed on the turntable **70** can change responsive to the turntable **70** rotating relative to the base assembly **12**. As another example, the video zones produced by the cameras **460** disposed on the chassis **54** can be maintained responsive to the position of the chassis **54** being maintained as the turntable **70** rotates relative to the base assembly **12**.

[0201] FIG. **31** is a perspective view of the electric boom **10**, according to an exemplary embodiment. FIG. **31** depicts an example of a first moveable element (e.g., the lift assembly **14**) having been moved relative to a second moveable element (e.g., the base assembly **12**). A first zone **3005** (e.g., a proximity zone), produced by a sensor **455** disposed on the lift assembly **14**, is shown to have changed with respect to a similar zone **3005** depicted in FIG. **30**.

[0202] FIG. **32** is a perspective view of the electric boom **10**, according to an exemplary embodiment. FIG. **32** depicts an example of the first moveable element (e.g., the lift assembly **14**) having been moved again relative to the second moveable element (e.g., the base assembly **12**). The first zone **3005** (e.g., the proximity zone), produced by the sensor **455** disposed on the lift assembly **14**, is shown to have changed with respect to the similar zone **3005** depicted in FIG. **31**. The first zone **3005** has been changed as the lift assembly **14** has been raised (e.g., moved relative to the base assembly **12**) from a first height, in FIG. **30**, to a second height, in FIG. **31**, and is now shown to be at a third height.

[0203] FIG. **33** is a perspective view of the electric boom **10**, according to an exemplary embodiment. The zones **2005** are shown to be capturing the lift assembly **14**, the platform assembly **16**, turntable **70**, and the base assembly **12**. The zones **2005** can be changed responsive to a first moveable element moving relative to a second moveable element.

[0204] FIG. **34** is a perspective view of the electric boom **10**, according to an exemplary embodiment. The sensors **455** and the cameras **460** can produce at least one **3405**. The zones **3405** can be and/or include the zones **3005**. The zones **3405** can be and/or include proximity zones and/or video zones. The zones **3405** can capture the area around the electric boom **10**. FIG. **34** depicts an example of objects disposed with the zones **3405**. While the zones **3405** can change responsive to a first moveable element moving relative to a second moveable element, the sensors **455** and/or the cameras **460** can be placed (e.g., disposed) on given portions of the electric boom **10** and/or the moveable elements thereof resulting in at least one zone **3405** being maintained responsive to a first moveable element moving relative to a second moveable element. For example, a first zone **2105** (e.g., a video zone) produced by a camera **460** disposed on a rearward portion of the chassis **54** can be maintained responsive to the turntable **70** moving relative to the base assembly **12**. The position of the chassis **54** can be maintained as the turntable **70** moves relative to the base assembly **12** and as a result a position in space of the camera **460** disposed on the chassis **54** can be maintained. The maintaining of the position of the camera **460** can result in the first zone **2105** being maintained.

[0205] FIG. **35** is a perspective view of the electric boom **10**, according to an exemplary embodiment. FIG. **35** depicts an example of a first moveable element (e.g., the lift assembly **14**) having been moved relative to a second moveable element (e.g., the base assembly **12**). A first zone **3405** (e.g., a proximity zone), produced by a sensor **455** disposed on the lift assembly **14**, is shown to have changed with respect to a similar zone **3405** depicted in FIG. **34**. Additionally, a second zone **3405** (e.g., a video zone), produced by a camera **460** disposed on the chassis **54**, is shown to

have been maintained with respect to a similar zone **3405** depicted in FIG. **34**. The maintaining of the second zone **3405** shows that even as the first moveable element has been moved an object proximate to the electric boom **10** is still in view of the second zone **3405**.

[0206] FIG. **36** is a perspective view of the electric boom **10**, according to an exemplary embodiment. FIG. **36** depicts an example of the first moveable element (e.g., the lift assembly **14**) having been moved again relative to the second moveable element (e.g., the base assembly **12**). The first zone **3405** (e.g., the proximity zone), produced by the sensor **455** disposed on the lift assembly **14**, is shown to have changed with respect to the similar zone **3405** depicted in FIG. **35**. The first zone **3405** has been changed again as the lift assembly **14** has been raised (e.g., moved relative to the base assembly **12**) from a first height, in FIG. **34**, to a second height, in FIG. **34**, and is now shown to be at a third height. Additionally, the second zone **3405** is shown to have been maintained and the maintaining of the second zone **3405** results in the second zone capturing the object proximate to the electric boom **10**.

[0207] FIG. **37** is a perspective view of the electric boom **10**, according to an exemplary embodiment. FIG. **37** depicts an example of the first moveable element (e.g., the lift assembly **14**) having been moved again relative to the second moveable element (e.g., the base assembly **12**). The first zone **3405** (e.g., the proximity zone), produced by the sensor **455** disposed on the lift assembly **14**, is shown to have changed with respect to the similar zone **3405** depicted in FIG. **36**. The first zone **2505** has been changed again as the lift assembly **14** has been raised (e.g., moved relative to the base assembly **12**) from a first height, in FIG. **34**, to a second height, in FIG. **35**, to a third height, in FIG. **36**, and is now shown to be at a fourth height. Additionally, the second zone **3405** is shown to have been maintained and the maintaining of the second zone **3405** results in the second zone capturing the object proximate to the electric boom **10**.

[0208] FIG. **38** is a perspective view of the electric boom **10**, according to an exemplary embodiment. As shown in FIG. **38**, the camera **460** can be disposed between a first tractive element **82** and a second tractive element **82**. The camera **460** can also be disposed between the turntable **70** and a ground surface. For example, the camera **460** can be positioned to capture an area beneath at least a portion of the turntable **70**.

[0209] FIG. **39** is a perspective view of the electric boom **10**, according to an exemplary embodiment. The counterweight **605** can define at least one recess **3905**. For example, the counterweight **605** can include and/or provide one or more indents. In some embodiments, the cameras **460** can be disposed and/or located at least partially within the recesses **3905**. The recesses **3905** can position the cameras **460** in one or more directions. For example, the recesses **3905** can position the cameras **460** in a downward direction.

Configuration of Exemplary Embodiments

[0210] The present disclosure contemplates methods, systems, and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine

properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0211] As utilized herein, the terms “approximately”, “about”, “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the invention as recited in the appended claims.

[0212] It should be noted that the terms “exemplary” and “example” as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0213] The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent, etc.) or moveable (e.g., removable, releasable, etc.). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

[0214] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” “between,” etc.) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0215] Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, Z, X and Y, X and Z, Y and Z, or X, Y, and Z (i.e., any combination of X, Y, and Z). Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present, unless otherwise indicated.

[0216] It is important to note that the construction and arrangement of the systems as shown in the exemplary embodiments is illustrative only. Although only a few embodiments of the present disclosure have been described in detail, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts or elements. It should be noted that the elements and/or assemblies of the components described herein may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present inventions. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the preferred and

other exemplary embodiments without departing from scope of the present disclosure or from the spirit of the appended claim.

Claims

- 1.** A vehicle, comprising: a chassis; a turntable rotatably coupled to the chassis, the turntable including a counterweight defining a recess; a boom assembly having a proximal end portion coupled to the turntable and a distal end portion opposite the proximal end portion; a platform coupled to the distal end portion of the boom assembly and configured to support an operator, wherein an obstructed area rearward of the platform is obstructed from view from the platform by the turntable; a first camera coupled to the turntable and having a first field of view including at least a portion of the obstructed area; a second camera directly coupled to the counterweight and received within the recess, the second camera having a second field of view angularly offset from the first field of view, wherein the recess opens downward to permit the second field of view to extend downward and out of the recess; and a display coupled to the platform and configured to display image data from at least one of the first camera or the second camera.
- 2.** The vehicle of claim 1, wherein the recess is a second recess, and wherein the counterweight of the turntable at least partially defines a first recess that receives the first camera, a portion of the counterweight extending between the first recess and the second recess to separate the first recess from the second recess.
- 3.** The vehicle of claim 1, wherein the recess is a second recess, and wherein the counterweight of the turntable defines a first recess that receives the first camera.
- 4.** The vehicle of claim 3, wherein the boom assembly extends forward from the turntable, wherein the first field of view extends rearward from the first camera, and wherein the recess is positioned to expose the first camera from behind the counterweight and below the counterweight.
- 5.** The vehicle of claim 1, wherein the counterweight defines a top end of the recess such that the counterweight overhangs the second camera.
- 6.** The vehicle of claim 1, wherein the first camera is directly and fixedly coupled to the counterweight.
- 7.** The vehicle of claim 1, further comprising a controller configured to: receive a signal indicating a relative position of a first component of the vehicle in one of a plurality of quadrants; determine that at least one of the first camera or the second camera is associated with the one of the plurality of quadrants; and provide image data from the at least one of the first camera or the second camera to the display based on the determination.
- 8.** A vehicle, comprising: a chassis; a turntable rotatably coupled to the chassis and including a counterweight; a boom assembly having a proximal end portion coupled to the turntable and a distal end portion opposite the proximal end portion; a platform coupled to the distal end portion of the boom assembly and configured to support an operator, wherein an obstructed area rearward of the platform is obstructed from view from the platform by the turntable; a first camera directly coupled to the counterweight of the turntable and having a first field of view including at least a portion of the obstructed area; and a display coupled to the platform and configured to display image data from at least one of the first camera or a second camera; wherein the counterweight of the turntable defines a first recess that receives the first camera and a second recess that receives the second camera directly coupled to the counterweight, a portion of the counterweight extending between the first recess and the second recess to separate the first recess from the second recess, wherein the first recess and the second recess each open downward such that the first camera and the second camera are exposed from below the counterweight.
- 9.** The vehicle of claim 8, wherein the second camera is directly coupled to the counterweight and has a second field of view angularly offset from the first field of view.
- 10.** The vehicle of claim 8, wherein the boom assembly extends forward from the turntable,

wherein the first field of view extends rearward from the first camera, and wherein the first recess is positioned to expose the first camera from behind the counterweight and below the counterweight.

11. The vehicle of claim 8, further comprising a controller configured to: receive a signal indicating a relative position of a first component of the vehicle in one of a plurality of quadrants; determine at least one of the first camera or the second camera is associated with the one of the plurality of quadrants; and provide image data from the at least one of the first camera or the second camera to the display based on the determination.

12. The vehicle of claim 8, wherein the first camera and the second camera are each directly and fixedly coupled to the counterweight.

13. The vehicle of claim 8, wherein the counterweight overhangs the first recess and the second recess.

14. The vehicle of claim 8, wherein the counterweight overhangs the first camera and the second camera.

15. A vehicle, comprising: a chassis; a turntable rotatably coupled to the chassis, the turntable including a counterweight defining a recess; a boom assembly having a proximal end portion coupled to the turntable and a distal end portion opposite the proximal end portion; a platform coupled to the distal end portion of the boom assembly and configured to support an operator, wherein an obstructed area rearward of the platform is obstructed from view from the platform by the turntable; a first camera coupled to the turntable and having a first field of view including at least a portion of the obstructed area; a second camera directly coupled to the counterweight and received within the recess, the second camera having a second field of view angularly offset from the first field of view, wherein the recess opens downward to permit the second field of view to extend downward and out of the recess; a display coupled to the platform and configured to display image data from at least one of the first camera or the second camera; and a controller configured to: receive a signal indicating a relative position of a first component of the vehicle in one of a plurality of quadrants; determine that at least one of the first camera or the second camera is associated with the one of the plurality of quadrants; and provide image data from the at least one of the first camera or the second camera to the display based on the determination.
